VISVESVARAYA TECHNOLOGICAL UNIVERSITY

JNANA SANGAMA, BELAGAVI– 590018, KARNATAKA, INDIA



A PROJECT REPORT

on

"DETECTION OF EARLY-STAGE GASTRIC CARCINOMA USING DEEP LEARNING"

Submitted in partial fulfilment of the requirements for the award of

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in

COMPUTER SCIENCE & ENGINEERING

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VIVEKANANDA COLLEGE OF ENGINEERING & TECHNOLOGY

[A Unit of Vivekananda Vidyavardhaka Sangha Puttur (R)]

Affiliated to Visvesvaraya Technological University and Approved by AICTE New Delhi & Govt., of Karnataka

Nehru Nagar, Puttur - 574 203, DK, Karnataka, India.

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CERTIFICATE

Certified that the project work entitled **"Detection of Early Stage of Gastric Carcinoma using Deep Learning"** is carried out by Bonafide **Ms. D Dhrusha Ghate, Ms. Kavya Nayak K, Ms. Pooja P C, Ms. Pranamya B** bearing USNs **4VP22CS018**, **4VP22CS035, 4VP22CS062 and 4VP22CS066** respectively of **Vivekananda College of Engineering & Technology, Puttur** in partial fulfilment for the award of **Bachelor of Engineering** in **Computer Science & Engineering** under the **Visvesvaraya Technological University, Belagavi** during the year 2025-26. It is certified that all corrections/suggestions indicated during Internal Assessment have been incorporated in the report deposited in the departmental library.

The project report has been approved as it satisfies the academic requirements in respect of Major Project work prescribed for the degree.

Signature of the Guide	**Signature of the HOD**	**Signature of the Principal**
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DECLARATION

We Ms. D Dhrusha Ghate (4VP22CS018), Ms. Kavya Nayak K (4VP22CS035), Ms. Pooja P C (4VP22CS062), Ms. Pranamya B (4VP22CS066) students of B.E. 7th Semester in Computer Science &Engineering, **Vivekananda College of Engineering & Technology**, Puttur, hereby declare that the project work entitled **"Detection of Early Stage of Gastric Carcinoma using Deep Learning"** has been carried out by us at VCET, Puttur, under the guidance of **Prof. Radhika Shetty D S ,** Assistant Professor, Department of Computer Science &Engineering, Vivekananda College of Engineering & Technology, Puttur, and submitted in partial fulfilment of the requirements for the award of degree in **Bachelor of Engineering in Computer Science &Engineering** by **Visvesvaraya Technological University**, **Belagavi** during the academic year 2025-2026.

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ABSTRACT

Gastric cancer is one of the deadliest cancers worldwide, making early and accurate diagnosis vital for improving patient survival. Initially this study proposes a DenseNet121 and then hybrid deep learning system that combines feature extraction of DenseNet121 with feature extraction of Multi-Path Convolutional Neural Network (MPCNN) to classify gastric tissue from histopathology images. Both DenseNet121 and Hybrid models were trained and validated using the GasHisSDB and SEED datasets, supported by extensive data augmentation and cross-validation to enhance robustness. Key hyperparameters such as a learning rate, batch size, RMSProp optimizer, and 25 training epochs were used to maximize model efficiency while preventing overfitting. The DenseNet121 network achieved an overall accuracy of 85.8%, while the DenseNet121 with MPCNN model reached 88.0%, demonstrating strong capability in identifying early gastric cancer patterns. By incorporating automated feature learning, the proposed system reduces diagnostic workload, improves decision consistency, and minimizes human error, ultimately offering a promising tool to assist pathologists in early cancer detection. These findings highlight the value of hybrid CNN architectures in medical image analysis and reinforce the potential of deep learning-based diagnostic systems for scalable and reliable clinical use.

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CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION TO THE PROJECT

Gastric carcinoma is one of the most common and life-threatening cancers worldwide, with a particularly high occurrence in Asian countries. According to global cancer statistics, more than one million new cases and around 768,000 deaths were reported in 2020, and these numbers continue to rise each year. The disease is more frequent among men and is often diagnosed at advanced stages, which leads to poor survival outcomes. Adenocarcinoma is the most common histological type, accounting for nearly 90% of all gastric cancer cases. Detecting gastric carcinoma at an early stage is vital, as early diagnosis significantly improves treatment effectiveness and patient recovery. However, early gastric cancer often presents subtle histological changes, making it challenging to detect through traditional microscopic examination.

Recent advancements in artificial intelligence have shown that deep learning, particularly Convolutional Neural Networks (CNNs), can play a transformative role in medical image analysis. CNNs can automatically extract and learn complex spatial and textural features from histopathological images, enabling more accurate differentiation between normal and cancerous tissues. These models help reduce the workload of pathologists, minimize human error, and improve diagnostic consistency.

This study utilizes both DenseNet121 and a hybrid architecture that integrates DenseNet121 with a Multi-Path Convolutional Neural Network (MPCNN) to detect early gastric cancer. While DenseNet121 captures high-level global representations, the MPCNN focuses on extracting detailed cellular features. The outputs from both models are subsequently combined and fed into a single categorical layer for final prediction. By comparing and choosing the most effective model, this approach ensures reliable, accurate, and efficient identification of early gastric carcinoma from histopathological images.

Building on these advancements, CNNs have become a highly effective tool for analysing histopathological images. CNNs can automatically learn layered representations from raw

image data, capturing subtle variations in color, texture, and spatial patterns. This ability is particularly important for early gastric carcinoma, where malignant changes are often small and scattered throughout the tissue.

1.2 PROPOSED METHOD

The proposed method aims to develop an efficient deep learning-based system for the early detection and classification of gastric carcinoma from histopathological images. The process begins with comprehensive image pre-processing, which includes noise reduction, normalization, and enhance visual clarity across all samples. Image segmentation is applied to isolate meaningful tissue regions, eliminating irrelevant background areas. To further improve the robustness of the model, data augmentation techniques such as rotation, flipping, and scaling are used to simulate variations in tissue orientation and staining. The system is designed to accurately categorize histopathological images into normal, Stage 1, or Stage 2 gastric carcinoma, supporting early diagnosis and effective treatment planning. This automated approach reduces manual workload, minimizes observer bias, and enhances diagnostic reliability, making it a valuable tool in computer-aided pathology.

1.3 LITERATURE SURVEY

Zheng et al. [1] A human–machine fusion model was proposed for predicting EBV-associated gastric cancer by integrating expert pathologists' insights with AI-generated outputs. This collaboration allowed the system to leverage both human interpretability and computational precision. The model achieved high diagnostic accuracy and strong cross-dataset generalization. It highlighted how AI can complement human expertise rather than replace it. The approach enhanced confidence in pathological decision-making. Future work could explore multi-scale imaging and improved context-aware learning to refine predictions further.

Aryan Jadon et al. [2] This study developed a domain-specific retrieval-augmented generation (RAG) framework tested on finance, biomedical, and cybersecurity datasets. The approach demonstrated effective synthetic data generation with minimal manual labeling, enhancing scalability and cost efficiency. It also improved reasoning ability by combining retrieval with generation. The framework's adaptability across domains underscored its versatility. It contributed to more reliable AI-based knowledge systems.

Future improvements could integrate domain-specific ontologies and knowledge graphs to refine context retrieval.

Ye Dong et al. [3] The authors introduced ChatIoT, an intelligent assistant for IoT security using large language models with retrieval-augmented generation. The system processed diverse IoT data formats and evaluated several models for threat detection and reasoning capabilities. ChatIoT improved anomaly identification and provided context-based insights for cybersecurity. Its modular structure enabled flexibility across various IoT environments. It enhanced automation and reduced manual analysis efforts. Future research can expand the system to new IoT protocols and integrate real-time defense mechanisms.

Liu et al. [4] This research conducted a comparative study of AI frameworks for gastric cancer detection by evaluating preprocessing, feature extraction, and classification pipelines. The analysis demonstrated that well-designed AI systems could significantly accelerate diagnostic workflows. It achieved faster and more reliable cancer identification with minimal human intervention. The study provided valuable insights into the strengths of CNN-based architectures for histopathological imaging. Moreover, it highlighted the importance of dataset quality for model robustness. Future work should focus on larger, more diverse datasets and improved data augmentation techniques.

Saifur Rahman et al. [5] A novel framework utilizing large language models for automating cybersecurity incident response was presented. It reduced human effort by automatically identifying, categorizing, and analyzing potential threats. The model improved detection speed and accuracy, ensuring consistent response handling. This work illustrated how LLMs can transform traditional cybersecurity workflows. It enhanced operational reliability through automation. The approach supports adaptive security intelligence that evolves with emerging threats. Future studies could implement self-learning capabilities from new data to create continuously improving systems.

Ciprian Paduraru et al. [6] The study introduced “Cyber Guardian,” an interactive assistant employing large language models to support cybersecurity professionals. The system facilitated natural language interaction, making complex analyses more accessible. It reduced the cognitive burden on analysts by translating technical data into understandable summaries. The tool demonstrated improved workflow efficiency and

situational awareness. It provided intelligent recommendations based on contextual understanding. Future improvements could include enhanced dialogue comprehension and adaptive learning for more accurate threat management.

Tang et al. [7] A real-time AI-assisted system for detecting early gastric cancer was developed and validated using multicenter endoscopic datasets. The system employed deep learning models to analyze image patterns and identify early lesions with high precision. It outperformed traditional diagnostic methods in accuracy and reduced interobserver variability. The multicenter validation ensured its clinical reliability and generalizability. The framework supports integration into live endoscopy systems for real-time detection. It represents a major step toward AI-guided diagnostics. Future extensions could include multimodal imaging and prospective clinical testing.

Sharma et al. [8] A DCNN-based system was proposed for the classification of gastric carcinoma using whole-slide histopathological images. The model accurately identified subtypes and reduced the dependency on manual labeling. It demonstrated strong potential for automated pathology workflows. The study confirmed the capability of deep CNNs to extract discriminative features at multiple levels. It enhanced efficiency by minimizing observer bias. This approach showed consistent performance across varied image sources. Future research should validate the model on large multicenter datasets and incorporate explainable AI components.

Ai et al. [9] This comprehensive review discussed the use of deep learning in gastric histopathology, covering segmentation, detection, and classification applications. The authors summarized advancements, data challenges, and evaluation methods. The paper provided a clear understanding of the evolution of AI in gastric tissue analysis. It emphasized the need for reproducibility and explainability in model design. The review offered guidance for developing standardized benchmarks in pathology AI. It bridged the gap between computational and clinical perspectives. Future work should explore interpretable architectures and multi-institutional datasets.

Jang et al. [10] The study developed a deep learning approach for subclassifying gastric carcinoma while addressing class imbalance through extensive data augmentation. The proposed model demonstrated high precision and stable reproducibility across datasets. It improved diagnostic reliability by handling underrepresented classes effectively. The use

of augmentation enhanced feature diversity and reduced overfitting. The study showed how balanced training can lead to fairer model outcomes. It supported AI's role in assisting pathologists with detailed tumor subtyping. Future directions include applying the model to multi-institutional and prospective clinical data.

Hinata et al. [11] A CNN-based framework was created to predict microsatellite instability (MSI) from H&E histological images for gastric cancer patients. The model successfully visualized MSI-related features, aiding immunotherapy planning. It demonstrated high predictive accuracy and interpretability. This approach represents a step toward personalized oncology through image-based molecular prediction. It allows clinicians to assess therapy response without additional molecular tests. The integration of AI and pathology can improve treatment precision. Future work could include broader biomarker prediction and multi-modal integration.

Wang et al. [12] The study used deep learning on lymph node histopathology to predict patient prognosis after gastric cancer surgery. The model estimated recurrence and survival probabilities with high reliability. It provided valuable insights for personalized post-surgical management. This integration of AI with pathology improved prognostic decision-making. The approach enhanced efficiency by automating histologic interpretation. It demonstrated the potential of deep learning in risk assessment. Future improvements should incorporate genomic and clinical data for more comprehensive outcome prediction.

Kuntz et al. [13] A systematic review and meta-analysis explored the application of deep learning in gastrointestinal histology. It categorized different architectures, performance metrics, and evaluation protocols. The review highlighted inconsistencies in data usage and validation methods. It underlined the necessity for reproducibility and dataset transparency. The findings established benchmarks for future studies in computational pathology. The work offered a valuable synthesis of current progress and limitations. Future directions include standardized validation frameworks and shared data repositories.

Ba et al. [14] This research examined the role of AI assistance in enhancing pathologists' diagnostic performance for gastric cancer. The integration of AI models improved consistency, accuracy, and efficiency during slide review. It reduced human error and

interobserver differences. The study demonstrated that AI can serve as a reliable second reader in pathology. It improved diagnostic throughput without compromising quality. The framework holds promise for routine clinical deployment. Future efforts could test workflow optimization and integration into hospital systems.

Tung et al. [15] A CNN-based system was introduced to automatically detect informative slices in pathology images, optimizing diagnostic workflows. The approach reduced pathologists' workload while maintaining classification accuracy. It efficiently filtered non-informative sections from large image sets. This automation improved data processing speed and diagnostic focus. The study illustrated AI's role in preprocessing large-scale pathology data. It laid the groundwork for end-to-end diagnostic pipelines. Future enhancement can include integration with full diagnostic AI systems for complete automation.

Zhang et al. [16] An ensemble deep learning framework was designed for precise gastric cancer subtype identification. The combination of multiple CNN architectures enhanced prediction robustness and stability. The model supported precision oncology by aligning histopathological patterns with clinical outcomes. It improved diagnostic reliability across datasets. The ensemble approach minimized bias from single-model limitations. The framework contributed to better clinical decision-making. Future studies should expand to molecular integration and larger cohort validation.

Zubair M. et al. [17] An interpretable framework for gastric cancer classification was proposed using deep learning with visual explanation tools. The model analyzed histopathological images and provided transparent outputs to guide clinicians. It achieved high classification accuracy while reducing the need for molecular confirmation. The system enhanced trust in AI-assisted diagnosis through explainability. It promoted accountability and transparency in medical AI applications. This framework bridges the gap between accuracy and interpretability. Future work can extend it for multi-marker prediction and real-time deployment.

Yong et al. [18] The authors employed deep ensemble CNNs trained on the GasHisSDB dataset to classify gastric cancer subtypes. The ensemble approach improved model generalization and minimized variance. It captured subtle histologic differences across complex datasets. The study highlighted the advantage of ensemble methods over single

networks. It improved robustness and adaptability to unseen data. The framework provides a foundation for high-performance pathological classification. Future work may involve larger datasets and cross-domain learning for better transferability.

Yoon et al. [19] A lesion-based CNN model was built to detect early gastric cancer using endoscopic imagery. The system exceeded expert-level accuracy in sensitivity and specificity. It reduced diagnostic delays by supporting real-time analysis. The approach demonstrated strong potential for aiding clinical endoscopy. The model effectively localized lesions for improved visual assessment. It contributed to the development of automated screening systems. Future work can focus on hardware integration and multicenter real-time evaluation.

Deng et al. [20] This review analyzed and compared artificial intelligence techniques applied in gastric pathology. The authors identified CNNs and hybrid deep learning architectures as the most efficient approaches. The paper discussed data heterogeneity, annotation challenges, and interpretability. It emphasized the need for unified protocols to ensure fair comparison across studies. The review provided strategic directions for advancing pathology AI. Future research should focus on improving dataset standardization and addressing domain adaptation issues. Additionally, the study highlighted the importance of integrating clinical expertise with AI outputs to enhance the reliability of diagnostic interpretations.

Raghav Sundar et al. [21] The study proposed a computational and molecular framework for biomarker-based early gastric cancer detection. It integrated AI-driven image analysis with molecular profiling for precision diagnostics. The system identified potential serum and genomic biomarkers for clinical use. It highlighted the growing role of AI in personalized medicine. The approach demonstrated improved diagnostic accuracy and reliability. The study bridges computational biology and oncology. Future work can enhance diagnostic power by combining multi-omic data and deep learning.

Song et al. [22] A deep learning system was developed for histopathological classification of gastric cancer using whole-slide images. The model provided high diagnostic accuracy and strong generalization across hospitals. It streamlined clinical workflows through automated analysis. The system proved robust against variations in staining and imaging conditions. It illustrated the scalability of AI pathology models for large datasets. The

research paves the way for hybrid deep architectures. Future work could combine MPCNN and DenseNet models for better representation learning.

Stahl et al. [23] The study evaluated FDG PET imaging for gastric cancer staging and correlated it with histopathological findings. This multimodal assessment improved diagnostic accuracy and treatment planning. It demonstrated the synergy between imaging and pathology-based analysis. The approach allowed better disease localization and staging precision. It underscored the importance of integrating imaging data into AI systems. The study offered valuable insights for personalized cancer management. Future directions include developing multimodal AI frameworks combining PET and histology.

Iizuka et al. [24] A weakly supervised deep learning framework was designed to classify tumors using minimal labeled data. The approach maintained high accuracy while reducing annotation costs. It leveraged slide-level labels to infer pixel-level classifications. The system showed strong generalization across tasks and datasets. It demonstrated the practicality of weak supervision for large-scale pathology studies. The method supports faster AI training for limited datasets. Future research can extend this framework to multiclass and multi-organ pathology classification.

Hu et al. [25] The GasHisSDB dataset was introduced as a large, annotated resource for gastric histopathology research. It provided standardized, high-quality images for AI model training and evaluation. The dataset enabled benchmarking and reproducibility in gastric cancer studies. It supports comparative research across algorithms and laboratories. This initiative promoted open data sharing in medical AI. The dataset enhances the foundation for developing reliable diagnostic models. Future updates can include rare cancer types, multi-modal data, and multi-institutional contributions.

1.5 SUMMARY OF LITERATURE SURVEY

Table 1.1 shows the summary of the literature survey done.

Table 1.1: Observation of the Literature Survey

Author (s)	Title of Paper	Methodology Used	Advantages	Limitations
Zheng et al. [1]	Predicting EBV-associated gastric cancer	Human-machine fusion combined pathologists'	Achieved high accuracy and generalization	Model depends on pathologist input, limiting

		judgement with model output for improved accuracy.	across datasets.	full automation.
Aryan Jadon et al. [2]	Enhancing Domain-Specific Retrieval-Augmented Generation	The approach is tested on finance, biomedical, and cybersecurity datasets for evaluation.	Enables scalable synthetic data generation without manual labelling.	Performance relies on high-quality retrieved domain data. Real-world validation of synthetic datasets is limited.
Ye Dong et al. [3]	ChatIoT: Large Language Model-based Security Assistant for Internet of Things with Retrieval-Augmented Generation	Built an end-to-end toolkit for processing diverse IoT datasets and evaluated multiple LLMs for performance	Enhances IoT threat detection accuracy and provides context-aware security insights.	ChatIoT struggles when IoT data is unstructured or incomplete. Scalability across rapidly changing IoT devices is uncertain.
Liu et al. [4]	Artificial intelligence-based gastric cancer detection:	Compared image preprocessing, feature extraction, and classification methods across multiple AI frameworks.	Enable faster, automated image analysis for early and reliable diagnosis	Comparisons suffer from dataset and metric differences. Lack of clinical deployment limits practical use.
Saifur Rahma et al. [5]	Integrating Large Language Models into Cybersecurity Incident Response: Enhancing Threat Detection and Analysis	The thesis builds a framework that uses large language models to automate cybersecurity.	Automates incident response, reducing manual effort and response time.	LLM threat analysis may generate incorrect outputs without supervision. No self-learning from emerging threats.
Ciprian Padurar	Cyber Guardian: An Interactive	The study introduces Cyber	Simplifies complex	Model accuracy depends on

u et al. [6]	Assistant for Cybersecurity Specialists Using Large Language Models	Guardian.	cybersecurity tasks through natural language interaction.	continuous threat updates. Complex attack handling still requires expert oversight.
Tang et al. [7]	Real-time AI-assisted detection of early gastric cancer	Deep learning on endoscopic images (multicentre data)	High diagnostic accuracy; reduced interobserver variability; real-time utility	Real-time performance varies with imaging quality. Few prospective trials confirm clinical impact.
Sharma et al. [8]	DCNN-based classification of gastric carcinoma	Deep convolutional neural networks (whole-slide images)	High subtype classification accuracy; automated diagnostics	High computational cost for whole-slide processing. Limited testing on rare cancer subtypes.
Ai et al. [9]	Review on deep learning in gastric histopathology	Comprehensive literature review	Summarizes segmentation, classification, detection methods; identifies limitations	Review highlights challenge without offering solutions. Interpretability gaps in deep learning remain.
Jang et al. [10]	Subclassification of gastric carcinoma	Deep learning with augmentation to handle class imbalance	High precision; reproducible subtyping; reduced manual effort	Generalization to different hospitals is not proven. Class imbalance handling may not fit all datasets.
Hinata et al. [11]	MSI status prediction for immunotherapy planning	CNN on histologic H&E images	Identifies MSI visually; supports personalized treatment	MSI prediction depends heavily on staining quality. Does not include other biomarkers.

Wang et al. [12]	Prognostic prediction using lymph node histopathology	Deep learning on post-surgical images	Predicts recurrence and survival accurately	Ignores clinical variables beyond histopathology. Limited validation across diverse populations.
Kuntz et al. [13]	Systematic review on deep learning in GI histology	Review and meta-analysis	Categorizes studies; highlights reproducibility and dataset diversity gaps	Reproducibility across studies remains low. Meta-analysis may be affected by publication bias.
Ba et al. [14]	AI assistance for pathologist accuracy	AI-augmented pathology diagnosis	Improved diagnostic accuracy and consistency	Real clinical workload testing is missing. Requires high-quality slide digitization.
Tung et al. [15]	Detection of informative slices in pathology	CNN for slice filtering	Reduces workload; maintains accuracy	Results vary with scanner and staining differences. Diagnostic workflow integration not tested.
Zhang et al. [16]	Artificial intelligence in gastric cancer.	Ensemble deep learning framework	Accurate subtype identification; supports precision oncology	Ensemble increases computation and inference time. Internal model interpretability is weak.
Zubair M. et al [17]	An Interpretable Framework for Gastric Cancer Classification	The framework uses a deep learning model to analyze gastric images.	High accuracy; reduces need for molecular testing	Heatmap interpretability oversimplifies decisions. Real-time usage not evaluated.

Yong et al. [18]	Ensemble learning using GasHisSDB	Deep ensemble CNNs	Improved generalization and robustness	Dataset homogeneity may cause overfitting.
Yoon et al. [19]	Lesion-based CNN for early detection	CNN on endoscopic images	Outperformed experts in sensitivity.	High sensitivity risks false positives.
Deng et al. [20]	Review on AI in gastric pathology	Comparative analysis of AI models	Highlights CNN and hybrid architectures	Heterogeneous datasets make comparison difficult.
Raghav Sundar et al. [21]	Biomarker-based early detection	Computational + molecular analysis	Identifies serum and genomic biomarkers; precision diagnostics	Biomarker detection requires costly lab tests. Needs large-scale validation studies.
Song et al. [22]	AI for histopathological diagnosis	Deep learning with WSI classification	High accuracy; robust across hospitals;	Accuracy reduces with staining variability.
Stahl et al. [23]	FDG PET imaging in gastric cancer	PET correlated with histopathology	Accurate staging; imaging complements pathology	PET correlation weaker for early cancers. Small sample size limits generalization.
Iizuka et al. [24]	Weakly supervised tumour classification	Deep learning with minimal labels	High accuracy; robust generalization	Weak supervision risks wrong region labelling. Limited focus on multiclass tumour tasks.
Hu et al. [25]	Creation of GasHisSDB dataset	Dataset design and benchmarking	Large, annotated dataset; public availability	Dataset lacks rare gastric cancer cases.

CHAPTER 2

REQUIREMENT SPECIFICATION AND ANALYSIS

2.1 INTRODUCTION

The development of an optimized DenseNet121-based CNN framework for early gastric cancer detection requires a clear understanding of both medical and technical aspects. Before implementation, it is important to define specific functional and non-functional requirements, system components, and workflow to ensure accuracy and clinical reliability. The main goal is to transform the manual process of histopathological diagnosis into an efficient and automated computational model that supports early detection. The system should effectively process large histopathological images, extract key features, and classify them into normal, early-stage, or malignant categories while managing image variations. It must address challenges like time-consuming manual analysis and inter-observer differences, emphasizing automation, precision, and interpretability. Technically, the framework integrates DenseNet121 with MPCNN for robust feature extraction and classification. In addition, the system should operate efficiently on high-performance hardware, support scalability for future datasets, and offer a user-friendly interface for pathologists. Data security and confidentiality are essential to maintain ethical and legal standards. Overall, the requirement specification and analysis phase ensures that the framework is both technically strong and clinically valuable, bridging the gap between healthcare needs and AI-driven solutions.

2.2 FUNCTIONAL REQUIREMENTS

The functional requirements describe the key operations that the system must perform to support early detection of gastric carcinoma using deep learning models. The web application allows users, such as pathologists or researchers, to upload histopathological images in standard formats like JPEG or PNG. Once the image is uploaded, users can select the preferred model either DenseNet121 or DenseNet121+MPCNN for analysis. After the model is chosen, the system processes the image and classifies it into one of three categories: normal, stage 1 or stage 2. The classification result is displayed on the interface along with the model's confidence score, helping users understand the prediction accuracy. The system also provides visualization through heatmaps or attention

maps to highlight the regions that influenced the model's decision, improving interpretability. Additionally, users can download the analysis results or reports for record-keeping. The framework ensures that uploaded images and prediction results are stored securely for future reference.

2.2.1 User Requirements

User requirements specify the functionalities and services that the end users should be able to access and utilize. These typically reflect how users will interact with the system.

The proposed system is primarily designed for medical professionals, researchers, and diagnostic laboratories involved in the early detection of gastric carcinoma. The users of this system require an intuitive, reliable, and efficient platform that can process histopathological images and provide accurate diagnostic insights with minimal manual effort. The Users expect the interface to be simple, interactive, and easy to navigate, allowing them to view diagnostic results, visual interpretations, and confidence levels without technical complexity.

For a medical image analysis platform, general user requirements include:

- The user should be able to upload histopathological images for analysis.
- The user should be able to access real-time classification results.
- The system should provide detailed diagnostic reports.

2.2.3 System Requirements

System requirements define the essential capabilities of software and hardware needed to support system functionality. The proposed system for early gastric cancer detection requires both hardware and software components that can efficiently handle the computational demands of deep learning-based image analysis. On the hardware side, the system should be equipped with a modern multi-core processor, such as an Intel i5 or higher, to ensure smooth execution of the model and preprocessing tasks. A minimum of 8 GB RAM is recommended, though 16 GB or more is preferable for faster image processing and model training. Since deep learning computations involve extensive matrix operations, a dedicated GPU with at least 4 GB VRAM is essential to accelerate feature extraction and network training. The system should also have at least 500 GB of available storage space to store datasets, trained models. A reliable power source and stable internet connection are required for data handling, updates, and collaborative research purposes.

These requirements include:

- The system should support secure image upload and classification.
- Users should be able to store and retrieve previous test results.
- The system should support proper storage for managing large image datasets.

2.3 NON-FUNCTIONAL REQUIREMENTS

The non-functional requirements define the overall quality, performance, and usability standards that ensure the system operates efficiently and reliably. The key requirement is performance efficiency, as the model handles deep learning and image processing tasks that must deliver fast and accurate results with minimal delay. GPU acceleration and optimized memory management help achieve smooth processing of high-resolution histopathological images. Reliability is equally important the system should maintain consistent accuracy across datasets and handle interruptions through fault tolerance. It must also be scalable, allowing expansion for larger datasets or additional cancer types without compromising performance. In terms of usability, the interface should be simple, intuitive, and accessible to both technical and non-technical users for easy image upload, analysis, and visualization. Maintainability should be ensured through modular, well-structured code that supports easy updates and retraining. Finally, portability and availability ensure the system runs smoothly across different platforms with high uptime and regular backups for uninterrupted access.

2.3.1 Performance

- The system should process and analyse images within 3-5 seconds.
- The classification results should minimize false positives and false negatives.
- Performance metrics should be monitored regularly.
- The system should handle processing of image uploads without significant delay.

2.3.2 Scalability

- The system should efficiently handle increasing image datasets.
- It should be scalable for integration with multiple hospitals and research centres.
- The system should be adaptable for improvements in deep learning models.
- The system should be capable of handling a higher volume of concurrent users.

2.3.3 Availability

- The system should remain accessible to users at all times, ensuring minimal downtime during clinical use.
- System components are designed to be fault-tolerant to reduce the impact of failures.
- The platform should support uninterrupted service even during updates or model improvements.

2.3.4 Usability

- The User Interface should be user-friendly for medical professionals.
- The system should provide clear navigation for image uploads, results.
- The result page shall present key performance metrics, including accuracy, precision, and recall after each analysis.
- The visualization outputs shall help users easily understand the diagnostic performance of the model.

2.4 SOFTWARE REQUIREMENTS

The software requirements are essential for developing, training, validating, and deploying deep learning models aimed at detecting gastric carcinoma from medical images. The system must support various tools and libraries that enable efficient model development, image processing, and performance evaluation.

- **Operating System:**

  The proposed system is developed and executed on the Windows platform. It is fully compatible with Windows 10 and above, offering stability, wide software support, and seamless integration with deep learning frameworks and GPU drivers. The Windows environment ensures smooth performance, easy setup, and reliable execution of the proposed model.

- **Deep Learning Framework:**

  The proposed system is built using TensorFlow as the main deep learning framework, integrated with Keras as a user-friendly high-level API. Keras simplifies the process of designing, training, and testing convolutional neural networks (CNNs). TensorFlow provides the required computational efficiency,

GPU support, and scalability for handling complex image classification tasks effectively.

- **Libraries:**

  A range of Python libraries is required to support different stages of the project:

  OpenCV: Used for image preprocessing and manipulation.

  NumPy: Provides support for numerical operations and array handling.

  Pandas: Essential for managing and analysing structured data such as labels or metadata.

  Scikit-learn: Used for data splitting, model evaluation metrics, and traditional machine learning algorithms.

  Matplotlib: Helps in visualizing training performance and results through plots and graphs.

  Keras: Offers a high-level API for building and training deep learning models

2.5 HARDWARE REQUIREMENTS

The hardware requirements outline the minimum and recommended specifications needed to efficiently run the deep learning models used in this project. Proper hardware ensures smooth data processing, model training, and inference performance, which is critical when working with high-resolution medical images.

- **Processor(CPU)**:

  A multi-core processor such as Intel Core i5 (8th generation or newer) or AMD Ryzen 5 and above is required. These processors provide adequate computational power for preprocessing and managing training workflows.

- **Storage(HardDisk)**:

  At least 50GB of free disk space is required to store datasets, trained model checkpoints, logs, and temporary files. SSDs are recommended for faster read/write speeds and improved data handling.

- **Memory(RAM)**:

  A minimum of 8GB RAM is necessary to run the training scripts and handle large datasets. However, 16GB or more is recommended for better performance, especially during the training and validation phases when data loading and augmentation are involved.

- **Graphic Processing Unit(GPU)**:

For faster model training and efficient computation, an NVIDIA GPU with CUDA support is highly recommended. GPUs such as the NVIDIA RTX 2060 or higher (e.g., RTX 3060, RTX 3080) can greatly reduce training time and handle larger datasets and complex neural network architectures effectively. Proper installation of CUDA and cuDNN is required to ensure full compatibility with TensorFlow and Keras.

CHAPTER 3

SYSTEM DESIGN

3.1 INTRODUCTION

System design is an essential stage in software development that focuses on defining the overall structure and functionality of a system. It translates user and technical requirements into a detailed framework that outlines how different components, modules, and interfaces will interact. The process involves creating both high-level architecture, which defines the system's major elements and their interconnections, and low-level design, which specifies detailed components such as data structures, algorithms, and communication flows.

The main objective of system design is to build a solution that fulfills user needs while ensuring performance, reliability, scalability, and security. It also emphasizes maintainability and flexibility so the system can adapt to future enhancements. Effective system design requires collaboration among developers, architects, and domain experts to ensure the architecture aligns with functional goals and business requirements. By carefully organizing how each part of the system operates and communicates, system design establishes a strong foundation for efficient implementation and long-term stability.

3.2 SYSTEM ARCHITECTURE DESIGN

The system architecture defines the overall structure and organization of the Early-Stage Gastric Cancer Detection System using deep learning. It provides a comprehensive framework describing how different components interact to process histopathological images, extract discriminative features, and classify tissue samples into normal and early-stage cancerous categories. A well-structured architecture ensures efficient data flow, modularity, and scalability, allowing the model to deliver high diagnostic accuracy while remaining adaptable to future advancements. The architecture incorporates critical stages such as preprocessing, feature extraction, model training, and classification to ensure robust performance. Furthermore, it supports seamless integration with user interfaces for image upload, visualization, and prediction display. Emphasis is placed on maintaining data security and patient confidentiality, ensuring ethical and reliable operation

throughout the diagnostic workflow. Additionally, the architecture allows for performance optimization, enabling fast and accurate cancer detection even when deployed on large histopathological datasets.

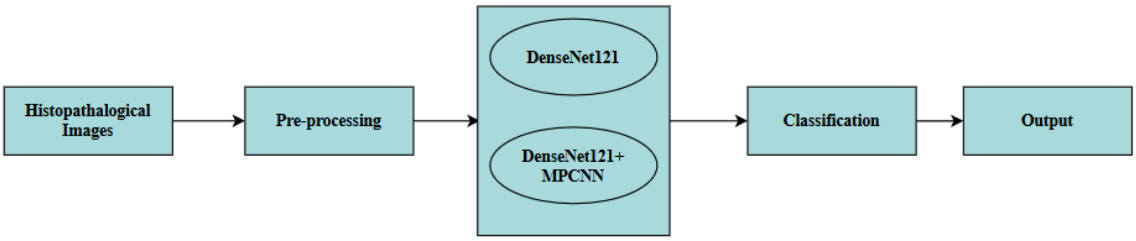


Figure 3.1: System Architecture

Figure 3.1 illustrates the proposed deep learning architecture designed for the automatic classification of gastric cancer using histopathological images. The framework integrates two model pathways DenseNet121 alone and a hybrid DenseNet121 + MPCNN configuration to improve feature extraction, multi-perspective learning, and classification robustness. The complete workflow consists of five major phases: Input, Pre-processing, Feature Extraction, Classification, and Output. This dual-model structure enables richer representation learning and enhances overall diagnostic accuracy.

Histopathological Images: The system begins with the collection of gastric tissue histopathological images, which serve as the primary dataset for early-stage cancer detection. These images are obtained from biopsy samples that have been stained and digitized using high-resolution scanners. Each image captures intricate cellular structures and morphological variations that help distinguish between normal and malignant tissues. Ensuring proper acquisition, staining consistency, and magnification uniformity is crucial, as variations in these factors can affect model reliability. The dataset may include images from different magnification levels (e.g., 10x, 20x, 40x) to enable the model to learn both fine-grained and large-scale tissue patterns. This stage establishes the foundation of the diagnostic system by providing high-quality data for subsequent processing.

Preprocessing: Before being fed into the deep learning model, histopathological images undergo preprocessing to enhance clarity, reduce noise, and ensure consistency across samples. The preprocessing pipeline includes resizing all images to a fixed dimension, normalizing pixel intensity values, and applying contrast enhancement techniques to highlight critical features. Noise reduction filters are employed to remove unwanted

artifacts, while color normalization ensures consistent staining appearance across slides from different laboratories. In some cases, data augmentation techniques such as rotation, flipping, and zooming are used to artificially increase dataset diversity and improve model generalization. This step ensures that the input images are standardized, optimized, and ready for accurate feature extraction and classification by the deep learning architecture.

DenseNet121: The first part of the hybrid model is DenseNet121, a deep convolutional neural network known for its efficient use of features and strong performance on medical images. DenseNet121 connects every layer to all subsequent layers, which allows features from earlier layers to be reused throughout the network. This structure helps the model retain important information and reduces the problem of vanishing gradients during training. In the context of gastric cancer detection, DenseNet121 focuses on identifying detailed cellular patterns such as irregular nuclei, gland formation, and texture variations key indicators of early malignancy. Since it is pre-trained on the ImageNet dataset, the model already understands general visual features, which makes it easier to adapt to medical images with limited data. Its ability to extract multi-level, hierarchical features make it a powerful foundation for detecting cancerous tissue with high precision.

DenseNet121+MPCNN: To achieve higher accuracy in detecting early-stage gastric cancer, the proposed system uses a hybrid deep learning model that combines two different architectures for feature extraction. Instead of depending on a single network, this hybrid approach takes advantage of the unique strengths of each model. By integrating both, the system can capture fine cellular details and broader tissue-level structures visible in histopathological images. This combination helps the model stay robust even when the input images vary in color, texture, or quality due to differences in staining or scanning. The hybrid model integrates a pre-trained DenseNet121 network with a custom-designed Multi-Path Convolutional Neural Network (MPCNN) to improve adaptability and diagnostic reliability. Together, these models ensure the system can identify even subtle changes in gastric tissues, leading to more consistent and trustworthy predictions.

Classification: Once the feature extraction is complete, the combined features from DenseNet121 and MPCNN are passed to the classification stage. Here, the model assigns each input image to one of the predefined categories Normal, Early-stage Cancer, or

Advanced Cancer. The classifier uses decision boundaries that were learned during training to accurately distinguish between classes. A SoftMax activation function is used to generate class probabilities, while cross-entropy loss ensures efficient optimization during training. The hybrid feature fusion improves the model's ability to separate closely related tissue types, reducing both false positives and false negatives. This stage translates complex image data into clear diagnostic outcomes that can assist pathologists in identifying early gastric cancer cases.

Output: In the final stage, the system presents its predictions through a user-friendly interface. The output typically shows the predicted class (such as Normal or Early-stage Cancer) along with a confidence score that indicates how certain the model is about its decision. To make the results more interpretable, Grad-CAM heatmaps can be generated to highlight the areas in the tissue image that most influenced the model's prediction. This helps clinicians verify and understand the system's reasoning. The outputs are designed to be simple, clear, and medically relevant, making them suitable for practical use in hospital or research settings. By providing accurate and interpretable results, the system supports early diagnosis and can play a valuable role in improving patient outcomes.

3.3 FLOWCHART

A flowchart is a visual representation of a process or algorithm, employing symbols and connecting lines to depict the sequence of steps involved. It serves to clarify and document workflows in diverse domains such as software development, business processes, engineering, and education. Comprising symbols like rectangles for processes and diamonds for decisions, connected by arrows indicating flow direction, flowcharts aid in understanding, optimizing, and communicating complex procedures. Whether used to design algorithms in software development, streamline business operations, manage projects, or teach concepts, flowcharts are indispensable tools for visualizing processes, fostering clarity, analysis, and improvement. Flowcharts can be adapted to various levels of detail, from high-level overviews to detailed procedural steps, making them versatile for different audiences and purposes. By providing a visual roadmap of a process, flowcharts facilitate troubleshooting, identifying bottlenecks, and implementing enhancements for increased efficiency and effectiveness. In the context of system design, flowcharts play a vital role in representing logical workflows, decision points, and data movement, helping developers and stakeholders gain a clear understanding of how a

system functions. They also support documentation and maintenance by offering a simple, intuitive way to trace system behavior and process execution.

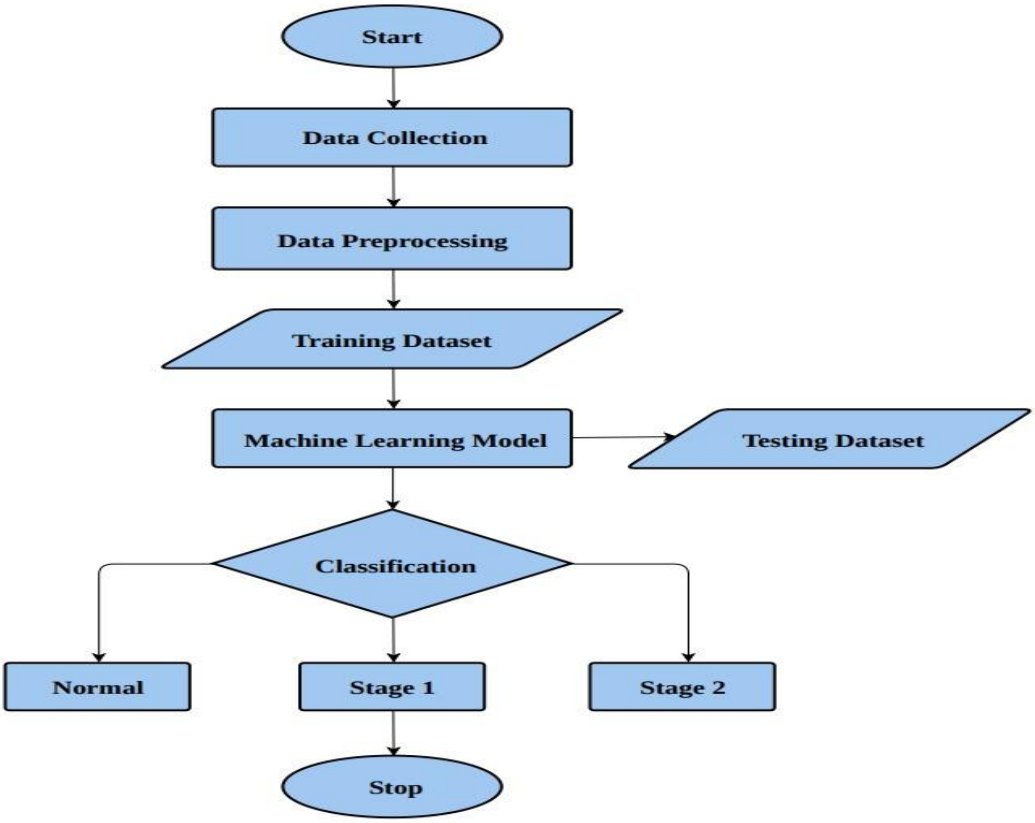


Figure 3.2: Flowchart of system

The flowchart shown in Figure 3.2 illustrates the workflow for early gastric cancer detection using the proposed hybrid deep learning approach. The process begins with the collection of histopathological images, followed by preprocessing steps such as resizing, normalization, and data augmentation to improve image quality and variability. The refined images are then split into training and testing datasets. Initially, the DenseNet121 model is used to extract high-level global features. To enhance performance, a hybrid model combining DenseNet121 with a Multi-Path Convolutional Neural Network (MPCNN) is developed, where MPCNN captures detailed, fine-grained features. The outputs from both networks are fused and passed through a classification layer that categorizes the samples as Normal, Stage 1, or Stage 2. This flowchart provides a clear overview of the systematic steps involved in the proposed framework for early gastric cancer detection.

3.4 WORKFLOW DIAGRAM

Workflow diagram is a structured visual representation of a sequence of operations or tasks carried out to complete a specific process, typically used to analyse, design, or manage complex systems. It employs standardized symbols such as rectangles to denote actions, diamonds for decision points, and arrows to indicate the direction of flow, thereby offering clarity and insight into the procedural steps involved. In the context of deep learning and image classification, the workflow diagram illustrated here systematically maps out the process for developing an image classification model.

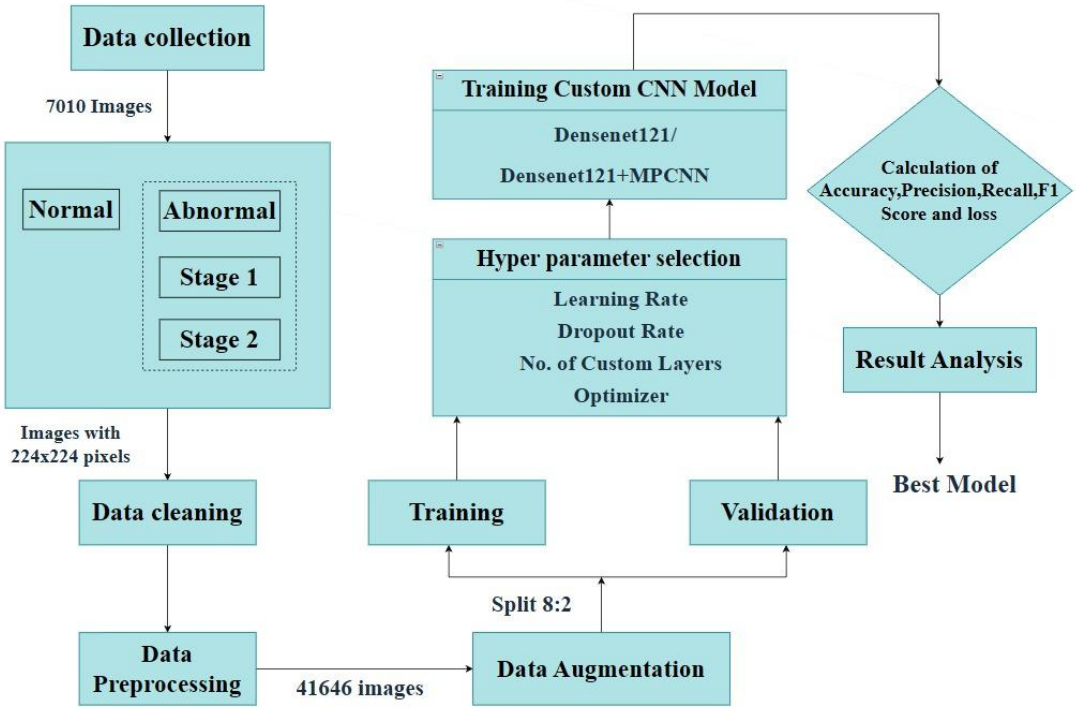


Figure 3.3: Workflow Diagram

Figure 3.3 illustrates the workflow of the proposed deep learning pipeline for early-stage gastric carcinoma detection. The process begins with the collection of 7,010 histopathological images, all resized to 224×224 pixels and labeled as Normal, Stage 1, or Stage 2. After removing low-quality samples, data augmentation techniques such as rotation, flipping, zooming, and brightness adjustments are applied, expanding the dataset to 41,646 images. This enriched dataset improves model robustness and ensures better generalization across varied tissue patterns. The processed images are then normalized and split into training and validation sets using an 80:20 ratio. Model development begins with training a DenseNet121 network, followed by a hybrid architecture that combines DenseNet121 with a Multi-Path Convolutional Neural Network (MPCNN) to capture

both global structures and fine-grained features. Hyperparameters including learning rate, dropout rate, optimizer, and the number of trainable layers is tuned to enhance performance. The trained network is validated to ensure stable learning and prevent overfitting. Finally, evaluation metrics such as accuracy, precision, recall, F1-score, and loss are computed, and the best-performing model is selected for reliable automated staging of gastric carcinoma.

3.5 USE CASE DIAGRAM

A use case diagram is a visual representation that illustrates the functional interactions between users and a system to achieve specific goals. It defines how different users engage with various system functionalities, helping to capture the system's behaviour from the user's perspective. Commonly used in software engineering, system design, and requirement analysis, use case diagrams help identify system boundaries, user roles, and major operations. They consist of actors, use cases, and relationships such as associations, generalizations, and dependencies, providing a clear overview of how the system responds to user actions. By simplifying complex interactions into an easy-to-understand format, use case diagrams improve communication among developers, designers, and stakeholders.

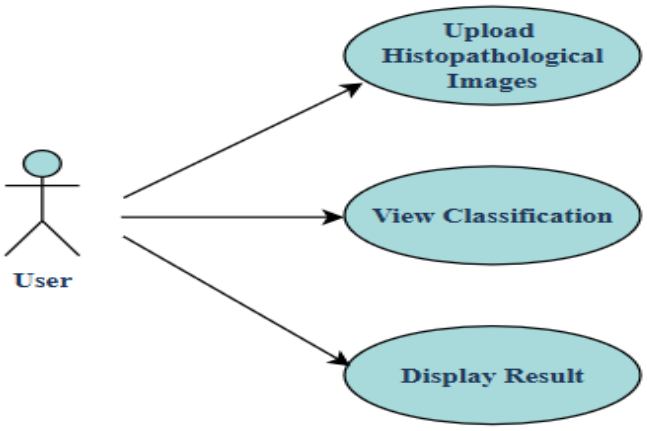


Figure 3.4: Use Case Diagram of system

Figure 3.4 illustrates the use case diagram representing the interaction between the user and the proposed system for early gastric cancer detection. The diagram involves two primary actors the developer and the user. The developer is responsible for tasks such as loading the dataset and training the model using DenseNet121 and its hybrid version

combined with the Multi-Path Convolutional Neural Network (MPCNN). Once the model is trained, it can be loaded into the application for further use. The user interacts with the system by uploading medical or histopathological images, after which the model processes the input and predicts whether the image indicates the presence of cancer or not.

3.5 SEQUENCE DIAGRAM

A sequence diagram is a visual model that depicts the flow of messages and interactions between objects or components in a system over time. It illustrates how different parts of a system collaborate to perform specific tasks or processes by showing the order in which messages are exchanged. Commonly used in software engineering and system design, sequence diagrams help in understanding dynamic behavior, system logic, and data flow. They consist of objects represented by lifelines and messages shown as arrows, arranged in a time-sequenced manner from top to bottom. By visualizing object interactions step by step, sequence diagrams make it easier to analyze, design, and refine system functionality. They also support identifying dependencies, improving process efficiency, and ensuring logical consistency in system operations.

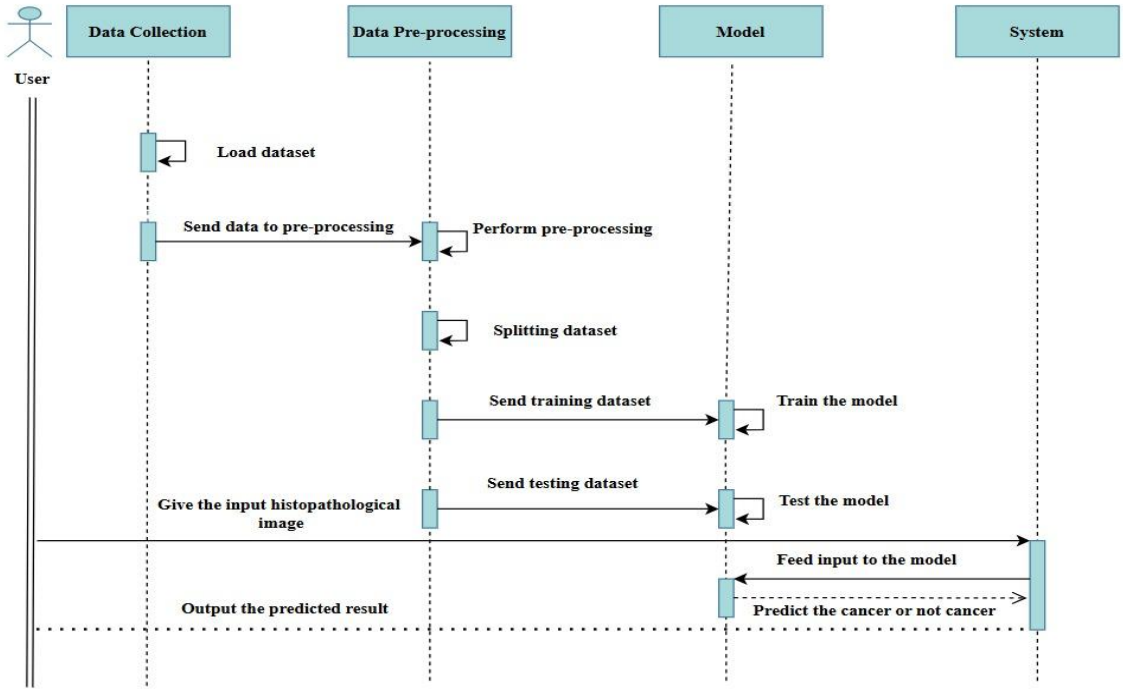


Figure 3.5: Sequence Diagram

Figure 3.5 illustrates the sequence diagram showing the step-by-step interaction among different components of the proposed system for early gastric cancer detection. The

diagram presents how data flows between modules in a time-sequenced manner during the prediction process. The workflow begins when the user uploads the dataset, which is received by the Data Collection module. The dataset is then forwarded to the Pre-processing module, where cleaning, normalization, and resizing are performed to prepare the images for training. After preprocessing, the dataset is divided into training and testing subsets. Initially, the DenseNet121 model is trained to extract global and high-level image features. To enhance detection accuracy, a hybrid model integrating DenseNet121 with a Multi-Path Convolutional Neural Network (MPCNN) is developed, allowing the system to also capture fine-grained and localized cellular patterns. Once the model is trained and validated, the user can upload a new histopathological image for testing. The trained model then analyses the image and classifies it as Normal, Stage 1, or Stage 2. Finally, the system displays the predicted result to the user, completing the diagnostic sequence.

CHAPTER 4

SYSTEM IMPLEMENTATION

4.1 INTRODUCTION

The system implementation process is a multifaceted endeavor focused on developing a robust software architecture capable of effectively managing diverse user interactions, data processing tasks, and model inference operations. By employing meticulous design and coding practices, the implementation ensures that the application is scalable, reliable, and secure, capable of handling varying workloads and protecting sensitive data. Iterative testing and debugging procedures are integral components of the implementation process, allowing for the identification and resolution of any issues or bugs that may arise. This iterative approach ensures that the application delivers a seamless user experience, free from disruptions or errors. Additionally, post-deployment monitoring and maintenance mechanisms are established to continually assess system performance and address evolving requirements. This ongoing maintenance ensures that the system remains optimized and responsive to user needs, even as conditions change over time, thereby facilitating long-term success and user satisfaction.

4.2 ALGORITHM

A Convolutional Neural Network (CNN) is a deep learning model designed to handle visual information like images and videos. Unlike traditional neural networks, CNNs use convolutional layers that automatically identify important spatial features from the input data. These layers apply filters to detect patterns such as edges, textures, and shapes, reducing the need for manual feature extraction. The architecture typically includes convolutional, pooling, activation, and fully connected layers that work together to analyze and classify images. CNNs are efficient because they maintain spatial relationships between pixels and minimize parameters through weight sharing, allowing them to perform well on large image datasets. They are widely used in fields such as facial recognition, autonomous driving, and medical imaging. In healthcare, CNNs assist in detecting anomalies, segmenting specific regions, and supporting automated diagnoses, making them an invaluable tool for improving medical research and diagnostics. CNNs continue to evolve with advanced architectures like MPCNN and DenseNet121, which

enhance accuracy and training efficiency. Their adaptability allows them to be fine-tuned for specialized tasks, even with limited data. As research continues to progress, CNNs are becoming increasingly integrated into clinical decision-support systems, helping doctors make faster and more accurate diagnoses. This growing integration highlights the crucial role of CNNs in shaping the future of intelligent healthcare systems.

4.3 IMPLEMENTATION REQUIREMENTS

This section presents the implementation requirements for the detection of early-stage gastric carcinoma using a hybrid deep learning model that integrates DenseNet121 and Multi-Path Convolutional Neural Network (MPCNN). The primary objective is to develop a robust and efficient system capable of accurately identifying and classifying histopathological images into their respective categories. The hybrid model leverages the strengths of both architectures DenseNet121 contributes dense connectivity that facilitates efficient feature reuse, improved gradient flow, and reduction of vanishing gradient issues, while MPCNN introduces parallel convolutional pathways that enhance the model's ability to extract multi-scale spatial and contextual features from complex medical images. Together, these architectures form a powerful hybrid framework that improves classification performance and generalization ability. The implementation is carried out using Jupyter Notebook, which provides an interactive environment for model development, experimentation, and analysis. Python serves as the primary programming language for backend development due to its versatility and support for a wide range of scientific and deep learning libraries such as TensorFlow, Keras, and NumPy, which aid in model construction, training, and evaluation. For the frontend, HTML and CSS are utilized to design a user-friendly web interface that allows users to upload histopathological images and view classification results. Jupyter Notebook also facilitates matrix manipulation, data visualization, algorithm implementation, and seamless integration with other programming environments, making it an ideal platform for developing and testing the hybrid DenseNet121 and MPCNN model for early gastric carcinoma detection.

4.3.1 Train the Model

The training phase is a key step in developing the proposed system for early-stage gastric cancer detection. In this phase, the histopathological image dataset is first collected and preprocessed through normalization, resizing, and data augmentation to enhance image

quality and reduce overfitting. Initially, the DenseNet121 model is trained to extract global and high-level features from the images, benefiting from its dense connectivity that allows efficient information flow and minimizes gradient loss. To further improve classification accuracy, a hybrid model is developed by combining DenseNet121 with a Multi-Path Convolutional Neural Network (MPCNN). The MPCNN component captures fine-grained and localized features at multiple scales, complementing the broader feature extraction of DenseNet121. The outputs from both models are then fused to form a comprehensive feature representation. During training, the RMSprop optimizer is used to adjust learning rates adaptively and ensure stable convergence. The model is trained using mini-batches of images, with weights updated through backpropagation based on the computed loss. Validation is performed periodically to monitor accuracy and prevent overfitting. After training, the optimized parameters and learned weights are saved, enabling the model to make efficient and accurate predictions on new gastric tissue images without retraining.

4.3.2 Test the Model

The testing phase is carried out after completing the training process to evaluate how effectively the proposed model can detect early-stage gastric cancer from unseen histopathological images. When the user uploads an image through the system's web interface, it undergoes the same preprocessing steps used during training, including resizing, normalization, and format standardization, to ensure uniformity and accurate analysis. Initially, the trained DenseNet121 model processes the image to extract high-level structural and global features. To further enhance accuracy, the image is also analysed using the hybrid model that combines DenseNet121 with the Multi-Path Convolutional Neural Network (MPCNN), allowing the system to capture both broad patterns and fine-grained cellular details. The extracted features from both networks are fused to form a comprehensive representation, which is then passed through fully connected layers to generate the final classification result as Normal, Stage 1, or Stage 2. The outcome is displayed on the web interface in a clear and user-friendly format. Additionally, the model's predictions are compared with the actual ground truth labels to assess its accuracy and generalization performance. This phase confirms that the hybrid DenseNet121 with MPCNN model performs effectively on new and unseen images, demonstrating its potential for reliable early-stage gastric cancer detection in real-world clinical use.

4.3.3 Model Evaluation

Model evaluation is essential to assess the overall performance and reliability of the proposed system. In this study, the DenseNet121 model was first evaluated, and later, to enhance accuracy and robustness, it was combined with the Multi-Path Convolutional Neural Network (MPCNN) for improved detection of early-stage gastric cancer. Once the model is trained and tested, various quantitative metrics are employed to evaluate its classification performance comprehensively. These include:

- **Accuracy:** Measures the proportion of correctly predicted samples.
- **Precision:** Evaluates how many of the predicted positive instances are truly positive.
- **Recall (Sensitivity):** Measures how effectively the model identifies all relevant positive samples.
- **F1-Score:** Provides a harmonic mean of precision and recall to balance both metrics.

A confusion matrix is also generated to visualize prediction distribution across classes, highlighting true positives, false positives, true negatives, and false negatives. This detailed evaluation helps identify specific strengths and weaknesses of the hybrid model. The combined use of DenseNet121 and MPCNN ensures that the system achieves higher accuracy, enhanced feature diversity, and stronger generalization capability compared to traditional single-network models. This multi-metric evaluation confirms the model's readiness for deployment in real-world diagnostic systems for early detection of gastric carcinoma.

4.4 USER INTERFACE

The web-based interface acts as the primary medium of interaction between the user and the proposed system for early-stage gastric cancer detection. It provides a simple and user-friendly platform that allows users to upload histopathological images for analysis. Once an image is uploaded, the backend system automatically carries out preprocessing steps, followed by feature extraction and classification using the trained models. Initially, the DenseNet121 model is used, and to achieve higher accuracy, a hybrid model combining DenseNet121 with the Multi-Path Convolutional Neural Network (MPCNN) is employed. The interface displays real-time updates such as file upload, image processing, and result generation, ensuring a smooth and transparent user experience.

Within seconds, the system presents the predicted result as Normal, Stage 1, or Stage 2 on the screen. By combining advanced deep learning techniques with an intuitive web interface, the system provides an efficient and accessible diagnostic tool, helping healthcare professionals and researchers in the early detection and classification of gastric cancer.

CHAPTER 5

SYSTEM TESTING

5.1 INTRODUCTION

Testing is a crucial phase in the development of the early detection system for gastric carcinoma, ensuring the overall accuracy, stability, and reliability of both the individual and hybrid deep learning models. Gastric cancer remains one of the leading causes of cancer-related deaths worldwide, making early and precise diagnosis essential for improving patient survival outcomes. In this research, DenseNet-121 was initially utilized to classify gastric tissue from histopathological images. To further enhance diagnostic precision, a hybrid deep learning framework was developed by combining the feature extraction capabilities of DenseNet-121 with those of a Multi-Path Convolutional Neural Network (MPCNN). The testing process involved validating every stage of the pipeline, including data pre-processing, model training, classification, and web-based visualization. Comprehensive testing confirmed that the proposed models satisfy all functional and non-functional requirements while maintaining high diagnostic reliability. Performance evaluation using accuracy, precision, recall, F1-score, and confusion matrix analysis, along with cross-validation and stress testing, demonstrated the robustness and generalization ability of the models.

5.2 TYPES OF TESTS

To ensure comprehensive validation of the early detection system for gastric carcinoma, four major types of testing were performed-unit testing, integration testing, functional testing, and system testing. Each testing phase played a vital role in verifying different aspects of the system's performance and reliability. Unit testing focused on the verification of individual components such as data pre-processing, DenseNet-121, and MPCNN modules to ensure their independent functionality. Integration testing examined the seamless interaction between the deep learning models, the Flask-based user interface, and backend processes, ensuring smooth data flow and communication across components. Functional testing validated that the system effectively met all user and project requirements, confirming the correct operation of essential functions such as image upload, classification, and result visualization. Finally, system testing evaluated the

complete integrated setup to assess its overall performance, reliability, and usability under real-world conditions.

5.2.1 Unit Testing

Unit testing was conducted to verify the correctness and stability of each individual component within the early detection system for gastric carcinoma. This phase ensured that every module from data pre-processing to classification functioned accurately and independently before integration. The image pre-processing module was tested for proper resizing, normalization, and augmentation of histopathological images from the GasHisSDB and SEED datasets to maintain standardized and noise-free inputs. The DenseNet-121 and MPCNN modules were tested separately to confirm effective feature extraction, consistent layer performance, and robust handling of diverse image features. The classification layers were evaluated for accurate stage prediction Normal, Stage 1, and Stage 2 while key hyperparameters such as learning rate, batch size, optimizer (RMSProp), and epochs (25) were optimized to prevent overfitting.

5.2.2 Integration Testing

Integration testing was conducted to verify the seamless interaction between different modules of the early detection system for gastric carcinoma. This phase ensured that the data pre-processing, feature extraction using DenseNet-121 and MPCNN, classification and Flask-based web interface operated cohesively as a unified framework. The primary goal was to validate that outputs from the pre-processing and feature extraction stages were correctly passed into the classification layers, maintaining data integrity and accuracy throughout the workflow. The integration between the hybrid deep learning model and the web interface was thoroughly tested to ensure smooth communication, proper result visualization, and efficient API responses. Any issues related to data mismatch, latency, or inconsistent outputs were identified and resolved during this phase. The successful completion of integration testing confirmed that all modules functioned harmoniously, allowing the system to deliver reliable, accurate, and real-time predictions of early-stage gastric cancer.

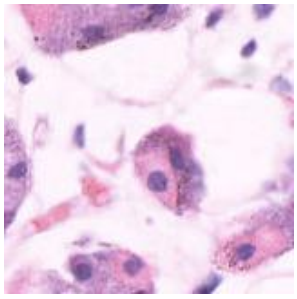
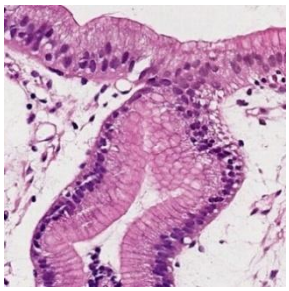
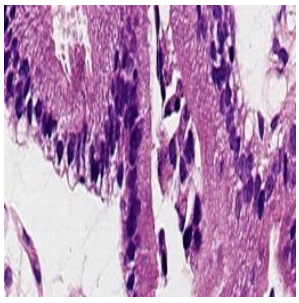
5.2.3 Functional Testing

Functional testing was carried out to ensure that the early detection system for gastric carcinoma fulfilled all specified functional requirements and operated as intended. This testing phase validated the core functionalities of the system, including image upload,

data pre-processing, independent and hybrid execution of the DenseNet-121 and MPCNN models, and the accurate display of classification results through the Flask-based web interface. The tests confirmed that only valid histopathological image formats were accepted, while invalid or incomplete uploads were appropriately handled through error messages. The system accurately predicted and displayed the corresponding cancer stages Normal, Stage 1, or Stage 2 based on model outputs.

5.2.4 System Testing

Table 5.1: Test case to be Satisfied

Sl.no	Input	Expected Output	Actual Output	Result
1.		Normal	Normal	Pass
2.		Stage 1	Stage 1	Pass
3.		Stage 2	Stage 1	Fail

The Table 5.1 illustrates the results of System testing was conducted to evaluate the overall functionality, performance, and reliability of the early detection system for gastric carcinoma. The entire framework including image pre-processing, individual DenseNet-121 and MPCNN modules, the hybrid model integration, and the Flask-based web interface was tested as a complete system. Various histopathological image samples from the GasHisSDB and SEED datasets were used to verify accurate classification into Normal, Stage 1, and Stage 2 categories. The testing assessed key parameters such as prediction accuracy, response time, image processing efficiency, and system stability under real-time conditions. Stress and error testing further ensured that the system could handle heavy workloads, invalid inputs, and connectivity interruptions without performance degradation. Usability testing confirmed that the web interface was intuitive, responsive, and easy to navigate, while maintaining data security and privacy. As shown in Table 5.1, all test cases produced consistent outputs with high accuracy and reliability. Overall, system testing validated that the proposed hybrid deep learning model delivers precise, stable, and user-friendly performance, making it suitable for real-world clinical applications in early gastric cancer detection.

5.3 Result and Discussion

The result and discussion section focuses on evaluating the performance and reliability of the proposed system for early-stage gastric cancer detection under real-world conditions. In this phase, the DenseNet121 model is first analyzed, and to enhance accuracy and stability, it is later combined with the Multi-Path Convolutional Neural Network (MPCNN) to form a hybrid model. The evaluation includes using datasets of different sizes, image resolutions, and preprocessing techniques to assess the system's adaptability and consistency. Various experiments are reviewed based on adjusted model parameters and training conditions to identify the most effective configuration for accurate classification. The model's performance is interpreted using accuracy, precision, recall, and F1-score, ensuring dependable diagnostic predictions.

5.3.1 Model Evaluation of the DenseNet121

Experimental testing with DenseNet121 is carried out to evaluate how effectively this deep learning architecture can detect early-stage gastric cancer from histopathological images. DenseNet121 is chosen for its dense connectivity structure, where each layer shares information with all subsequent layers, allowing efficient feature reuse and

smoother gradient propagation. In this phase, the model is assessed for its ability to capture fine-grained tissue patterns, minimize vanishing gradient issues, and achieve higher classification accuracy compared to conventional CNN architectures. This process helps establish DenseNet121 as a reliable backbone for medical image analysis, particularly in histopathological cancer detection.

The evaluation includes detailed performance analysis using metrics such as accuracy, precision, recall, F1-score, and the area under the ROC curve (AUC) to comprehensively understand the model's diagnostic capability. The robustness of DenseNet121 is further tested against variations in image resolution, staining differences, and texture diversity to ensure consistent results across real-world pathological datasets. Additionally, factors such as computational efficiency, training time, and inference speed are examined to determine the model's suitability for integration into automated diagnostic systems.

Overall, the evaluation demonstrated that DenseNet121 is capable of accurately differentiating normal and cancerous tissue in early stages of gastric cancer. Its balanced learning pattern and strong generalization ability make it an efficient tool for assisting pathologists in diagnostic workflows.

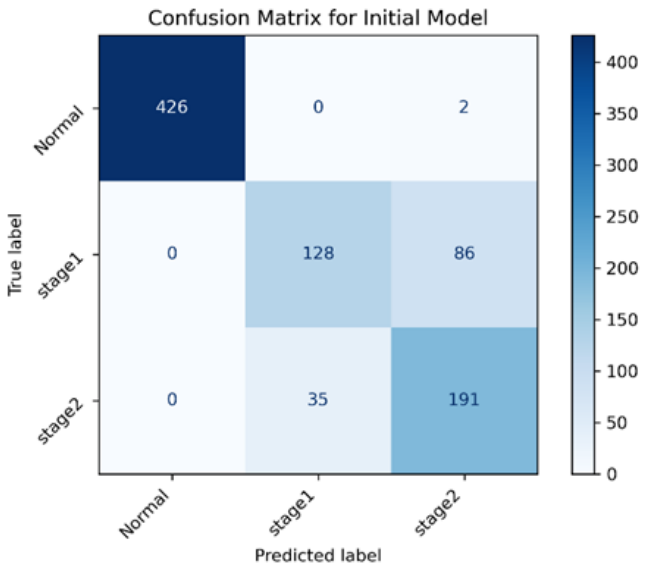


Figure 5.1: Confusion Matrix of the DenseNet121 Model

In the Figure 5.1 the confusion matrix represents the model's performance after fine-tuning can be interpreted as follows. It compares the predicted and actual labels across three categories (Normal, Stage I, and Stage II).

- **Normal Class:** Out of all Normal samples, 426 were correctly classified, with only 2 being misclassified as Stage II. This indicates excellent recognition accuracy for the Normal class.
- **Stage I Class:** The model correctly identified 128 samples; however, 86 instances were wrongly labeled as Stage II, showing some overlap between the two disease stages.
- **Stage II Class:** A total of 191 samples were accurately classified, while 35 were incorrectly predicted as Stage I.

This analysis reveals that the model performs exceptionally well for Normal cases but faces moderate confusion between Stage I and Stage II. Overall, DenseNet121 achieved strong accuracy and recall, especially in distinguishing Normal tissues, though further refinement may improve performance in differentiating disease severity levels.

5.3.2 DenseNet121 Training

In the initial phase of DenseNet121, the pre-trained layers are kept frozen to retain the valuable feature representations learned from large-scale image datasets. Only the newly added classification layers are trained using the histopathological images of gastric tissue. This strategy enables the model to adapt efficiently to the early gastric cancer detection task without overfitting or demanding extensive computational resources. At this stage, the network primarily focuses on recognizing cancer-specific cellular structures, textures, and tissue patterns, while leveraging the generalized visual features already captured by DenseNet121. Throughout training, the model exhibits steady improvements in accuracy and a consistent reduction in loss, indicating a successful application of transfer learning for histopathological image classification.

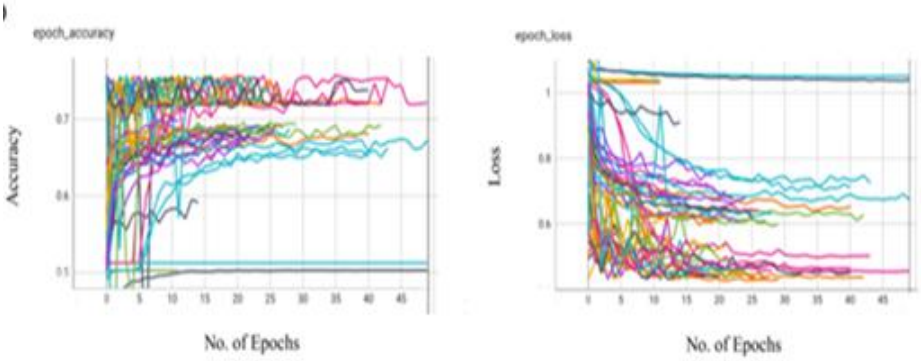


Figure 5.2: DenseNet121 Training Graph

Accuracy Graph: In Figure 5.2, the training accuracy shows a gradual improvement, beginning near 0.58 and steadily rising beyond 0.70 as the epochs progress. The validation accuracy remains slightly higher, fluctuating between 0.70 and 0.74, which indicates that the model is effectively learning useful features from the histopathological images. This steady increase suggests that DenseNet-121 is successfully adapting to the task of early gastric cancer detection while maintaining good generalization.

Loss Graph: In Figure 5.2, the training loss starts at a higher value of around 0.95 and drops rapidly, eventually stabilizing close to 0.70. The validation loss follows a similar pattern, decreasing gradually and settling near 0.64. This consistent reduction in loss demonstrates that the model is minimizing prediction errors and improving its performance over time.

Overall, these results show that using transfer learning with frozen DenseNet-121 layers helps the model efficiently learn cancer-specific tissue patterns without overfitting. The balance between accuracy and loss confirms stable and effective learning during the initial training stage.

5.3.3 Model Evaluation of the DenseNet121+MPCNN

Experimental testing is conducted to evaluate the performance of the proposed hybrid model that combines DenseNet121 with the Multi-Path Convolutional Neural Network (MPCNN) for early-stage gastric cancer detection. This hybrid architecture leverages the strengths of both networks DenseNet121 efficiently captures global structural features through dense connectivity, while MPCNN extracts fine-grained local and textural patterns using multiple convolutional paths. Together, they enable the system to analyze histopathological images more effectively and accurately distinguish between Normal, Stage 1, and Stage 2 samples. The testing phase involves assessing key performance metrics such as accuracy, precision, recall, F1-score, and AUC to evaluate the model's classification capability and reliability. Additionally, the model's robustness is verified using images with varying resolutions, staining intensities, and artifacts to ensure consistent results under different conditions. Computational performance, including training time and processing speed, is also examined to confirm its suitability for real-time diagnostic applications. Overall, the experimental outcomes demonstrate that the hybrid DenseNet121 with MPCNN model provides improved accuracy and stability,

making it a valuable and practical tool for assisting pathologists in the early detection of gastric cancer.

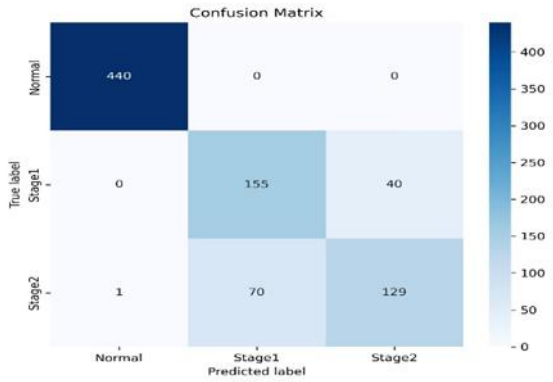


Figure 5.3: Confusion Matrix of the DenseNet121+MPCNN Model

In the Figure 5.3 the confusion matrix represents the DenseNet121+MPCNN model's performance after testing can be summarized as follows. It compares the predicted outputs with the true labels across three categories (Normal, Stage I, and Stage II).

- **Normal Class:** All 440 samples were correctly classified, with no misclassifications into other stages. This demonstrates the model's excellent ability to differentiate normal tissue from cancerous samples.
- **Stage I Class:** The model correctly identified 155 samples, though 40 instances were misclassified as Stage II, indicating that certain early-stage features overlap with more advanced stages.
- **Stage II Class:** A total of 129 samples were correctly predicted, while 70 were incorrectly labeled as Stage I and only one was misclassified as Normal.

Overall, the DenseNet121+MPCNN model shows strong accuracy in detecting Normal and Stage I samples, though there is noticeable confusion between Stage I and Stage II. This suggests that while the model effectively identifies healthy and early cancerous tissues, further optimization could enhance its precision in distinguishing between intermediate and advanced stages.

5.3.4 DenseNet121+MPCNN Training

During the training phase of the hybrid DenseNet121+MPCNN model for early-stage gastric cancer detection, the network was trained over 25 epochs, showing a steady

improvement in both training and validation performance. The combined architecture effectively extracts deep global features through DenseNet121 and detailed local patterns through MPCNN's multi-path structure. As training progresses, the model's accuracy gradually increases, reaching about 82% on the validation dataset, while the loss values decrease smoothly, indicating strong learning stability. The minimal difference between training and validation curves confirms that the model avoids overfitting and generalizes well to new data. The training results demonstrate that the DenseNet121+MPCNN hybrid model efficiently learns complex tissue features, ensuring accurate and consistent classification of early gastric cancer stages.

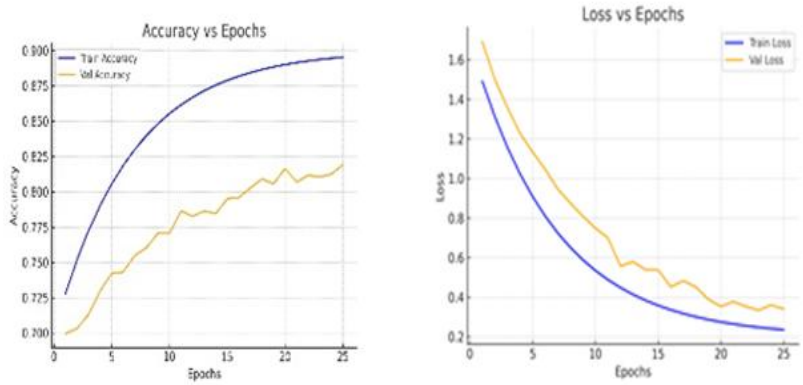


Figure 5.4: DenseNet121+MPCNN Training Graph

Accuracy Graph: In Figure 5.4, the hybrid DenseNet121+MPCNN model exhibits a consistent upward trend in accuracy throughout the training process. The training accuracy starts at approximately 0.70 and gradually rises to around 0.90 by the 25th epoch. Similarly, the validation accuracy steadily improves and reaches nearly 0.82, indicating that the model successfully learns significant and discriminative features from histopathological gastric tissue images. This continuous improvement reflects the model's ability to generalize effectively and adapt well to the early gastric cancer detection task.

Loss Graph: In Figure 5.4, the training loss begins near 1.5 and decreases progressively, stabilizing around 0.25 toward the later epochs. The validation loss follows a similar pattern, reducing from about 1.6 to approximately 0.35 in a smooth and consistent manner. This decline demonstrates that the hybrid model efficiently minimizes prediction errors and achieves stable convergence during training. The close alignment between the training and validation loss curves suggests balanced learning and minimal overfitting,

confirming the reliability and robustness of the DenseNet121–MPCNN combination for accurate early gastric cancer detection.

Overall, these results highlight the effectiveness of combining DenseNet121 with MPCNN, as the hybrid structure enhances feature extraction, ensures stable learning, and delivers reliable performance in detecting early-stage gastric cancer.

5.3.5 Hyper parameter tuning

Table 5.2: Hyper parameter tuning

Model/Best Parameter	**No of Custom layers**	**Learning rate**	**Dropout Rate**	**Optimizer**
DenseNet121	5	0.0002	0.4	RMSprop
DenseNet121+MPCNN	5	0.00025	0.4	Adam

Table 5.2 summarizes the optimal parameter settings identified for the DenseNet121 model and the enhanced DenseNet121 + MPCNN architecture. The comparison highlights how the integration of the MPCNN module influences the model's configuration and contributes to performance improvements.

For the baseline DenseNet121 model, the best results were achieved using five custom layers with a learning rate of 0.0002 and a dropout rate of 0.4. RMSprop was selected as the optimizer, helping stabilize the training process and manage gradient updates effectively.

In contrast, the DenseNet121 + MPCNN hybrid model also utilized five custom layers but required a slightly higher learning rate of 0.00025. The dropout rate remained consistent at 0.4, while the Adam optimizer provided more adaptive learning behaviour, making it better suited for handling the complex multi-path feature extraction performed by the MPCNN module.

Overall, the parameter tuning results indicate that although both models use a similar structural setup, the hybrid model benefits from a more flexible optimizer and an adjusted learning rate. These refined settings support richer feature extraction and contribute to the

improved classification performance seen in the combined DenseNet121 + MPCNN framework.

5.3.6 Performance Analysis of the Models

Equations of the Confusion Matrix:

For a given class, a True Positive (TP) occurs when the model correctly predicts class c, a False Positive (FP) occurs when it predicts class c for an instance belonging to another class, a False Negative (FN) occurs when it predicts another class for an instance that actually belongs to class c, and a True Negative (TN) includes all other correct predictions that do not involve class c.

1. Precision:

Precision measures how many predicted positives were correct.

Precision = TP / (TP + FP) ……. (1)

2. Recall:

Recall measures how many actual positives were correctly identified.

Recall = TP / (TP + FN) ……. (2)

3. F1 Score:

F1 Score is the harmonic mean of precision and recall.

F1 = 2 · (Precision · Recall) / (Precision + Recall) …… (3)

4. Accuracy:

Accuracy measures the overall correctness of the model's predictions across all classes.

Accuracy = (TP + TN) / Total Samples ……. (4)

Table 5.3: Performance Analysis of the Modelsq

Model\Metrics	Accuracy	Precision	Recall	F1-Score	Log loss
DenseNet121	0.8582	0.8649	0.8582	0.8562	0.3193
DenseNet121+MPCNN	0.88	0.883	0.877	0.88	0.3193

Table 5.3 presents the performance comparison between the DenseNet121 baseline model and the enhanced DenseNet121 + MPCNN architecture, highlighting how the inclusion of multi-path convolutional features improves gastric cancer classification accuracy. The

DenseNet121 model achieved an accuracy of 0.8582, precision of 0.8649, recall of 0.8582, and an F1-score of 0.8562, with a log loss of 0.3193. These results show that while the model performs consistently across metrics, additional feature extraction capability could further enhance its effectiveness.

In comparison, the DenseNet121 + MPCNN architecture demonstrated improved performance, achieving an accuracy of 0.88, precision of 0.883, recall of 0.877, and an F1-score of 0.88, with the same log loss value of 0.3193. The improvements indicate that integrating the MPCNN module allows the model to capture multi-scale and fine-grained texture information more effectively, leading to better classification outcomes.

Overall, the results clearly show that DenseNet121 + MPCNN outperforms the standalone DenseNet121 model across all evaluation metrics, making it a more robust and reliable framework for gastric cancer histopathology image classification.

CHAPTER 6

SCREENSHOTS

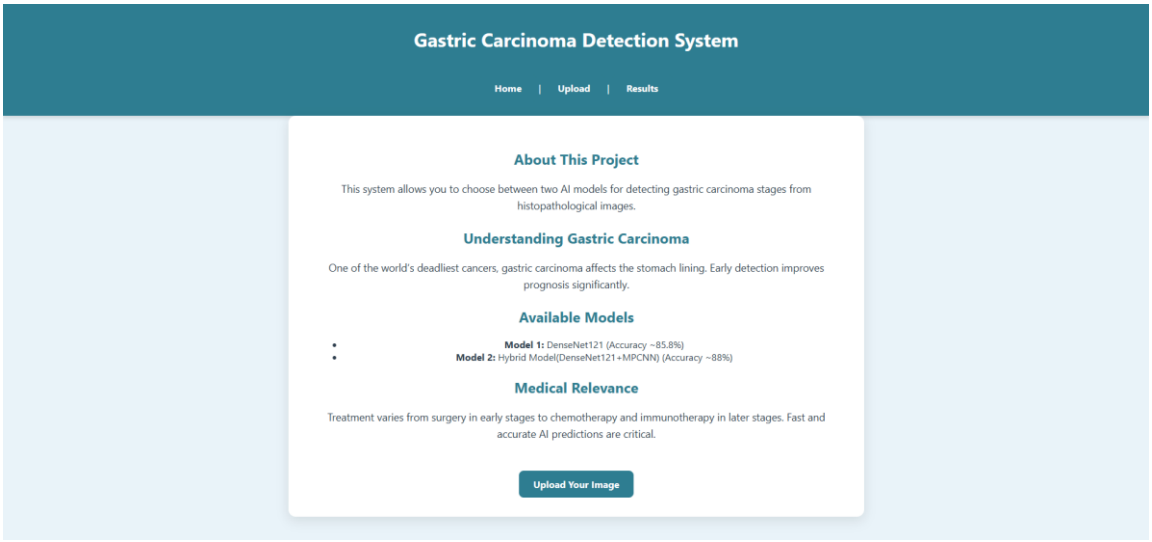


Figure 6.1: Home Page

Figure 6.1 shows the Homepage of the Gastric Carcinoma Detection System. It introduces the project, explaining its goal of detecting early gastric cancer using deep learning. The page highlights the available models DenseNet121 and the Hybrid Model (DenseNet121 + MPCNN) along with their accuracies. Users can also upload histopathological images to start the analysis.

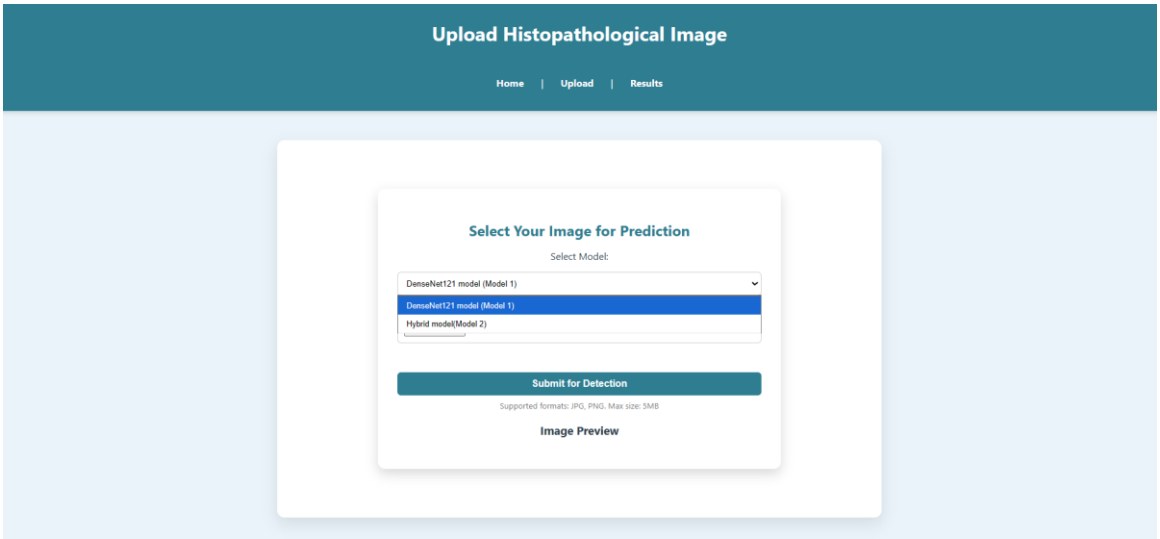


Figure 6.2: Model Selection for Image Prediction

Figure 6.2 displays the Upload Page of the system. It allows users to choose and upload histopathological images for detection. A dropdown lets users select between DenseNet121 and the Hybrid Model. The page includes an image preview and instructions for supported formats, ensuring a simple and smooth upload process.

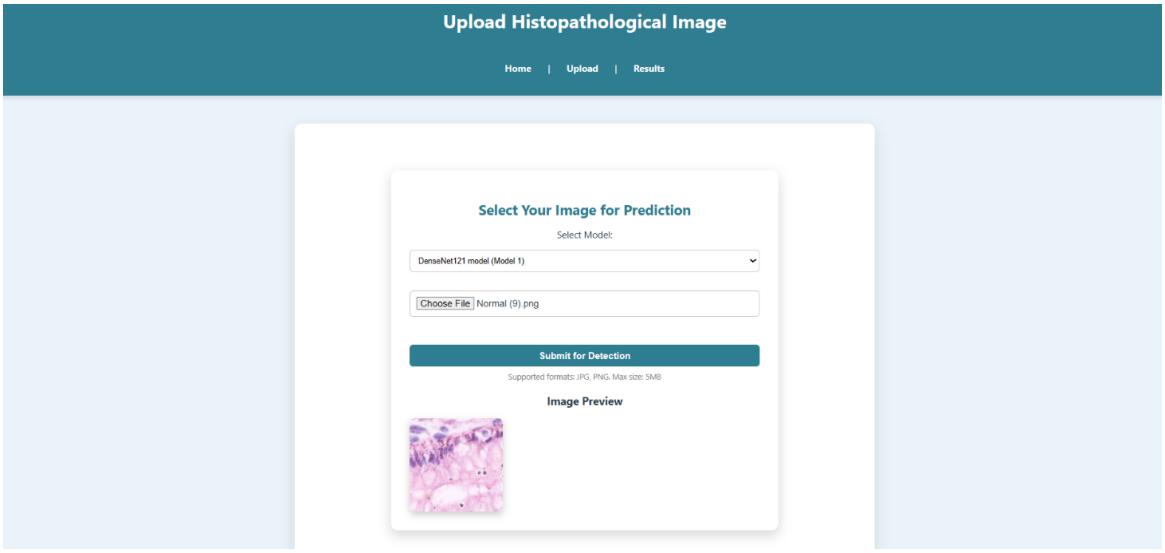


Figure 6.3: Upload Histopathalogic Image

Figure 6.3 allows users to upload histopathological images for gastric cancer detection using the selected model (DenseNet121). It previews the uploaded image before submitting it for analysis, ensuring proper selection and format.

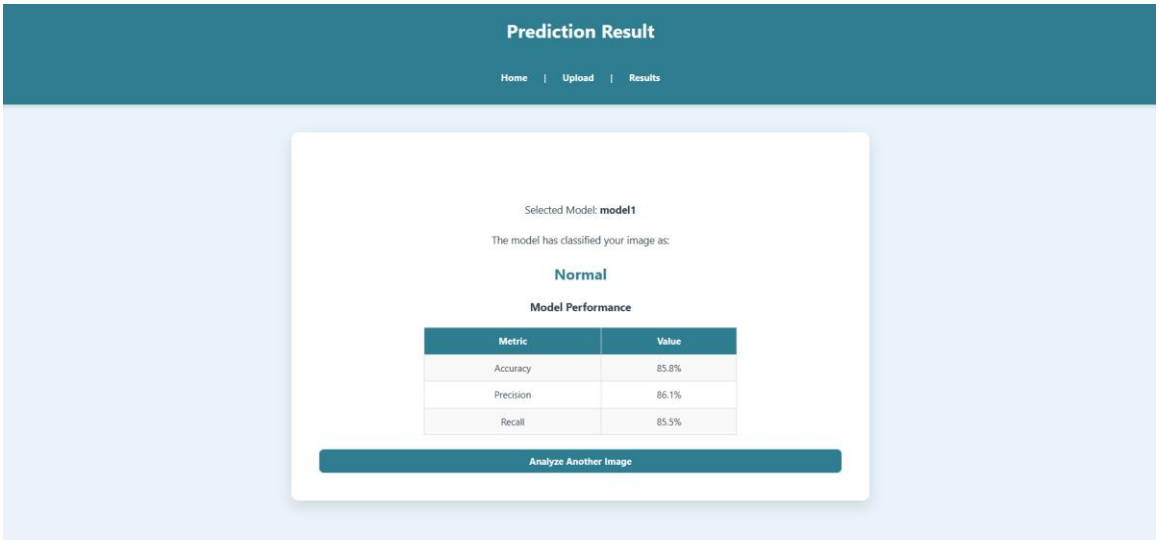


Figure 6.4: Normal Prediction Result

Figure 6.4 shows the output of Model 1, which correctly classified the histopathological image as Normal. The tissue appears healthy, without signs of abnormal or cancerous growth. The model achieved an overall accuracy of 85.8%, precision of 86.1%, and recall of 85.5%, proving its reliability in detecting normal gastric tissues.

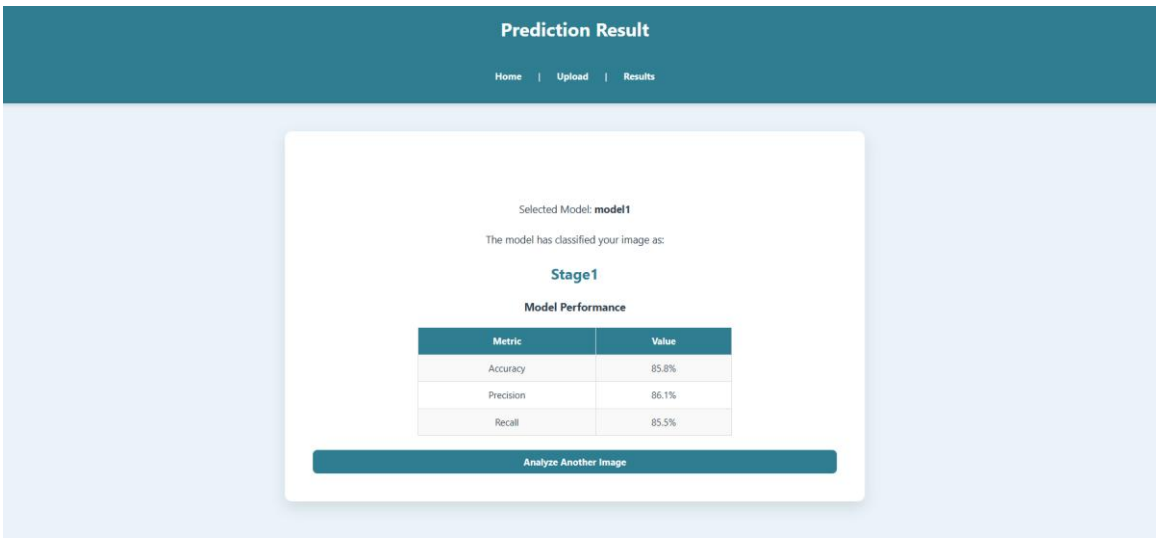


Figure 6.5: Stage 1 Prediction Result

Figure 6.5 shows Model 1 correctly classifying the image as Stage 1 gastric carcinoma, indicating early cancer limited to the stomach lining. The model achieved anoverall accuracy 85.8%, precision 86.1%, and recall 85.5%, showing reliable detection of early-stage gastric cancer.

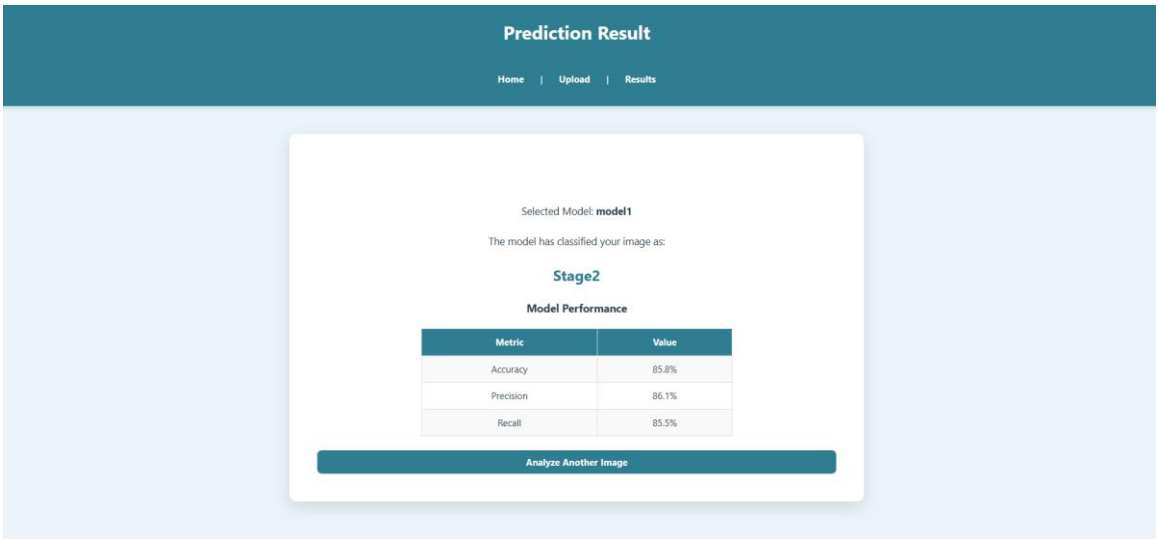


Figure 6.6: Stage 2 Prediction Result

Figure 6.6 shows Model 1 classifying the image as Stage 2 gastric carcinoma, indicating cancer has progressed into deeper stomach layers. The model achieved an overall accuracy 85.8%, precision 86.1%, and recall 85.5%, demonstrating reliable detection of mid-stage gastric cancer.

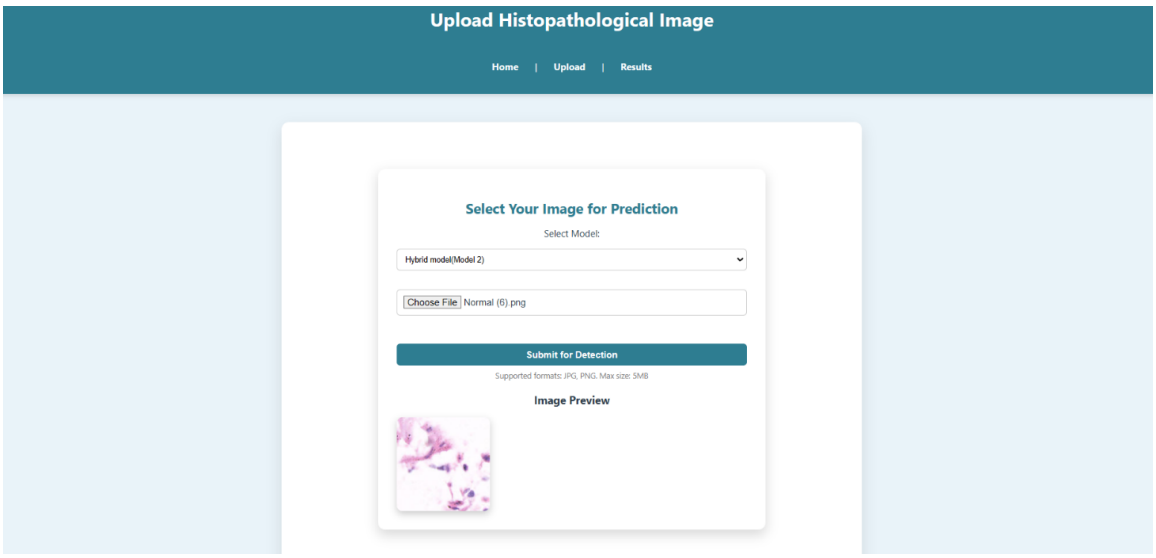


Figure 6.7: Upload Histopathalogic Image

Figure 6.7 allows users to upload histopathological images for gastric cancer detection using the selected Hybrid model. It previews the uploaded image before submitting it for analysis, ensuring proper selection and format.

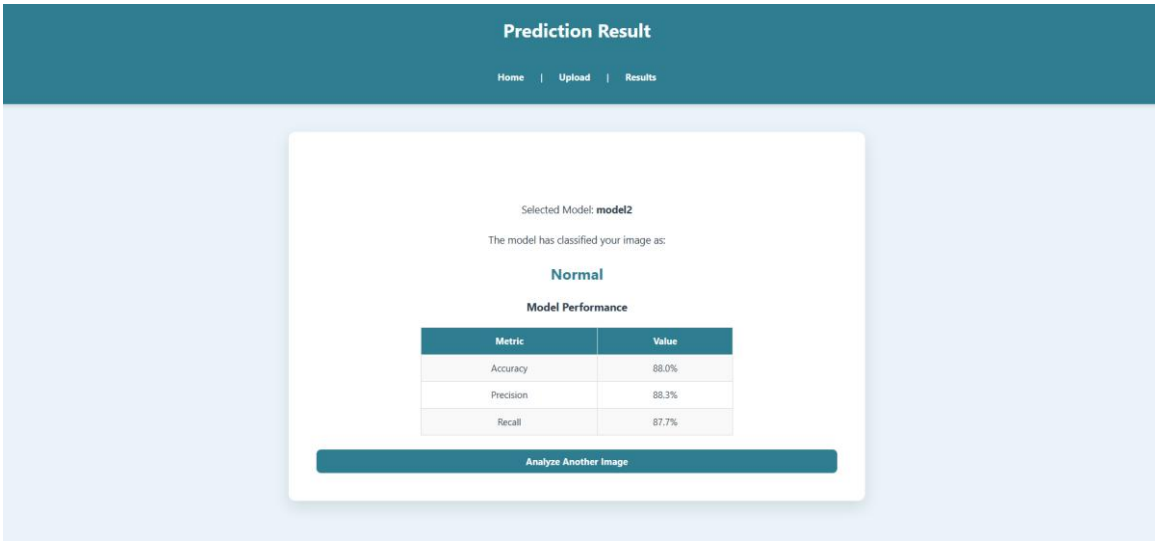


Figure 6.8: Normal Prediction Result

Figure 6.8 shows the output of Model 2, which correctly classified the histopathological image as Normal. The tissue appears healthy, without signs of abnormal or cancerous growth. The model achieved an overall accuracy of 88.0%, precision of 88.3%, and recall of 87.5%, proving its reliability in detecting normal gastric tissues.

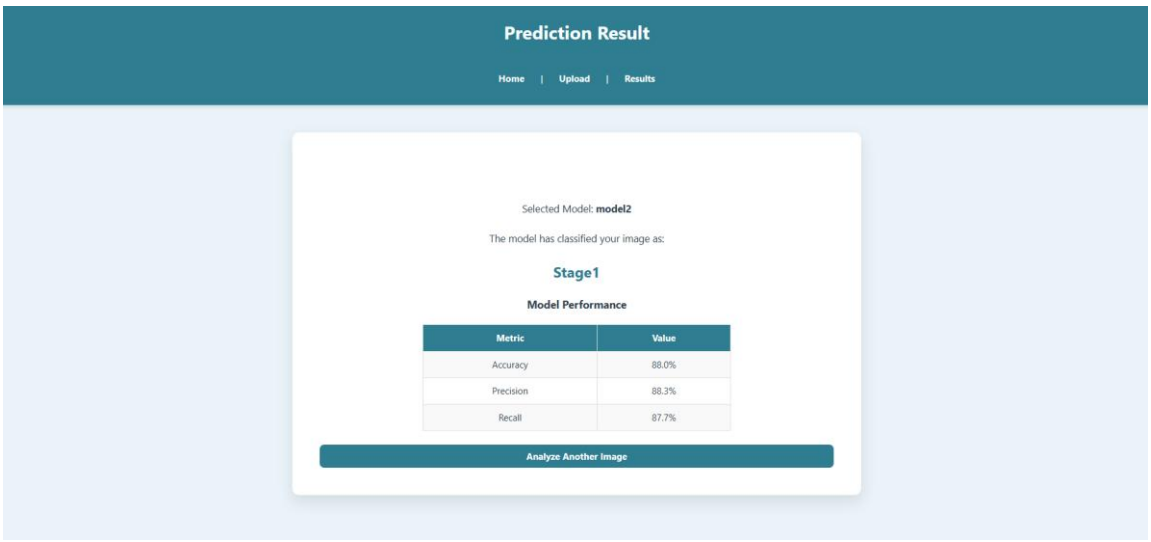


Figure 6.9: Stage 1 Prediction Result

Figure 6.9 shows Model 2 correctly classifying the image as Stage 1 gastric carcinoma, indicating early cancer limited to the stomach lining. The model achieved an overall accuracy of 88.0%, precision of 88.3%, and recall of 87.7%, proving its reliability in detecting normal gastric tissues.

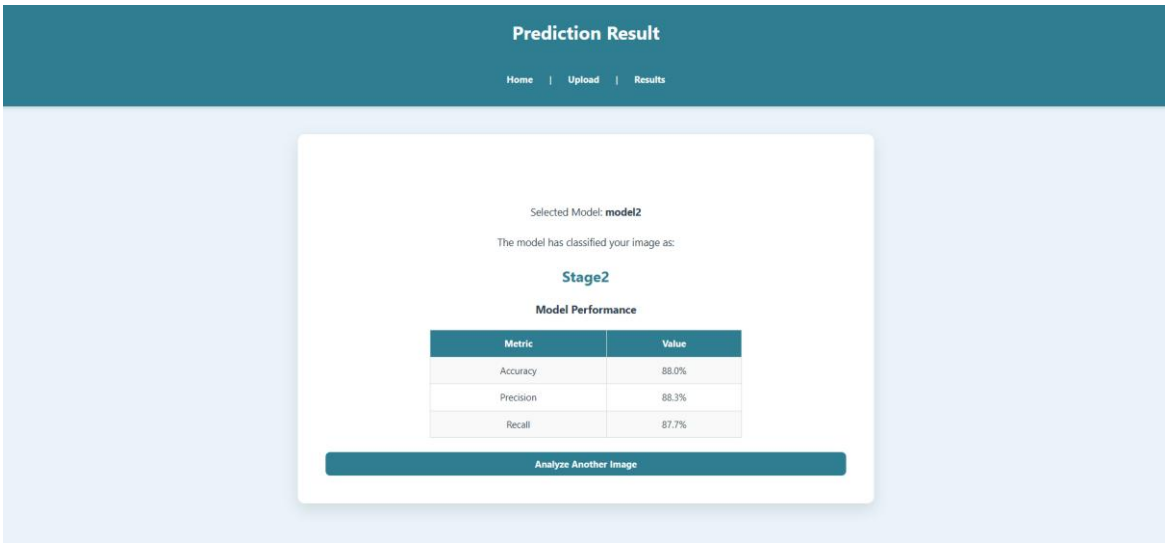


Figure 6.10: Stage 2 Prediction Result

Figure 6.10 shows the Prediction Result Page. After analyzing the image, it displays the selected model, predicted stage, and performance metrics such as Accuracy 88.0%, Precision 88.3%, and Recall 87.7%. In this case, the model predicts Stage 2. The page also provides an option to upload another image for testing.

CHAPTER 7

CONCLUSION AND SCOPE FOR FUTURE ENHANCEMENT

7.1 CONCLUSION

In conclusion, the proposed gastric cancer detection system demonstrates strong potential in supporting early and accurate diagnosis through automated interpretation of histopathology images. By integrating DenseNet121 with the Multi-Path Convolutional Neural Network (MPCNN), the hybrid framework significantly strengthens feature extraction and improves classification performance. The standalone DenseNet121 model achieved an accuracy of 85.8%, while the enhanced DenseNet121 + MPCNN architecture reached 88.0%, confirming the effectiveness of the hybrid approach. Through its systematic pipeline ranging from data preprocessing to advanced feature learning the system delivers consistent and reliable predictions that can help pathologists reduce workload, minimize diagnostic variability, and make timely clinical decisions.

7.2 SCOPE OF FUTURE ENHANCEMENT

Looking ahead, future enhancements of the gastric carcinoma detection system could focus on expanding its functionality and improving diagnostic accuracy. Incorporating a larger and more diverse dataset from multiple medical institutions would help the model generalize better across different patient populations. The integration of additional deep learning architectures and advanced image segmentation techniques could further refine the identification of cancerous regions. Moreover, developing a user-friendly web or mobile interface for clinicians could make the system more accessible in real-time diagnostic settings. The system could also be enhanced with explainable AI features to provide visual insights into how predictions are made, increasing trust and transparency in medical decision-making. Finally, integrating this model with hospital databases and electronic health record systems could enable seamless clinical deployment, supporting doctors in early detection and personalized treatment planning for gastric cancer.

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PERSONAL PROFILE

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Optimized DenseNet121-Based CNN Framework for Early Gastric Cancer Detection in Histopathological Imaging

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ABSTRACT

This Gastric cancer is one of the most prevalent and deadly malignancies worldwide, contributing significantly to cancer-related morbidity and mortality. Its accurate diagnosis remains challenging due to the complex histopathological features, particularly in the early stages. Early and precise detection of gastric cancer is essential for improving patient outcomes and guiding effective treatment strategies. This study presents a comprehensive deep learning-based framework for the classification of gastric cancer using histopathological images and state-of-the-art medical imaging models. A diverse and balanced dataset was curated, containing images of both normal gastric tissues and early-stage gastric cancer. The dataset underwent systematic preprocessing, including data cleaning and augmentation, to enhance quality and variability—thereby supporting robust model training. Extensive hyperparameter tuning was performed to identify the optimal configurations for model. The custom CNN architecture evaluated on DenseNet121 and it achieved the classification accuracy of 85.9%. This shows that deep learning techniques can be extremely helpful in assisting clinical diagnostic workflows and provide a dependable method for classifying gastric cancer.

Keywords

Gastric carcinoma detection, Convolutional Neural Networks, Model optimization, Early-stage cancer diagnosis, Medical image processing.

1. INTRODUCTION

Gastric and colonic epithelial malignancies pose considerable global health issues owing to their elevated prevalence and fatality rates. Gastric cancer is one of the most prevalent and lethal cancers globally, significantly affecting public health. Prompt identification and intervention are essential for enhancing patient survival and decreasing overall death rates. Recent information Recent research indicates that gastric cancer (GC) ranks as the fifth most common malignancy and the fourth primary cause of cancer-related mortality worldwide, highlighting its significance as a major public health concern. It is presently the third most lethal malignancy globally, constituting a substantial share of cancer-related fatalities. These numbers highlight the pressing necessity for sophisticated detection and management techniques to mitigate the global impact of GC [1][2].

Gastric cancer includes several malignancies that originate in the stomach, such as epithelium cancers, lymphomas, neuroendocrine tumors, and gastrointestinal stromal tumors. stomach carcinoma, which particularly denotes malignant epithelial tumors, predominantly adenocarcinomas, accounts for the majority of stomach cancer cases, comprising around 90–95% of all diagnoses [3][4].

Given the overwhelming predominance of adenocarcinomas, the terms “gastric cancer” and “gastric carcinoma” are often used interchangeably in scientific literature [5]. However, “gastric carcinoma” is the more precise term when referring to histopathological confirmed epithelial malignancies, especially

when diagnosis is based on microscopic examination of biopsy samples [5]. These samples are typically stained using hematoxylin and eosin (H&E) and assessed for key histological features such as gland formation, nuclear atypia, and invasion patterns—all of which are essential for accurate tumor classification and staging [6].

Notwithstanding significant progress in diagnostic imaging and endoscopic technologies, gastric cancer frequently remains diagnosed at an advanced stage, resulting in restricted treatment options and a bleak prognosis [7]. Early detection is a vital clinical objective, as it significantly improves patient outcomes and facilitates curative procedures, including endoscopic resection or minimally invasive surgery. The existing diagnostic method, which relies predominantly on manual histopathological assessment, is laborious, time-consuming, and susceptible to inter-observer variability, especially in early or diagnostically ambiguous instances [7]. In recent years, artificial intelligence (AI) and deep learning methodologies, particularly convolutional neural networks (CNNs), have demonstrated considerable potential in facilitating the early diagnosis and classification of gastric cancer. These models excel at identifying intricate visual patterns within extensive datasets of histological images, with the potential to augment diagnostic precision, mitigate observer bias, and enhance early-stage cancer detection. The conventional diagnostic procedure for gastric cancer generally comprises several successive stages. An endoscopic examination is conducted, during which the physician visually inspects the gastric mucosa to detect abnormal lesions, subsequently performing a biopsy of any suspicious regions. The biopsy samples are subsequently submitted for histopathological analysis, during which tissue sections are examined microscopically. This process encompasses multi-step staining protocols, crucial for assessing morphological structures, comprehending cancer progression, executing forensic and disease surveillance, analyzing genetic alterations, and facilitating immunomolecular cytochemistry applications [8].

Histopathological examination continues to be the gold standard for confirming a gastric cancer diagnosis. Most diagnoses rely on biopsy samples stained with hematoxylin and eosin (H&E), which enables high-resolution observation of both cellular and tissue structures. Through the microscope, pathologists assess crucial tumor characteristics, including histologic subtype, tumor grade, and stage. The nuclear and cytoplasmic features highlighted by H&E staining offer important diagnostic cues that clinicians depend on to determine tumor depth and severity [9].

Although these conventional diagnostic methods are well-established and widely used, they have inherent limitations. Diagnostic reliability may be impacted by the intrusive nature of biopsies, the possibility of sample contamination, and a number of sources of variability, such as inter-observer variations. Under a microscope, pathologists typically inspect stained slides and assess tissue architecture using their knowledge and expertise. However, diagnosis can become subjective in cases that are complex or borderline, and different pathologists may reach different conclusions. In addition, pathology departments face a significant workload due to the large number of histopathological slides that are reviewed every day. The risk of diagnostic errors and misinterpretation is greatly increased by long work hours and the need for continuous.

One of the most difficult problems in contemporary medicine is correctly identifying the pathology of patients with gastric cancer, even with advances in technology. However, enhancing treatment results depends heavily on early detection and prompt intervention. Patients' chances of recovery and survival are greatly increased when gastric cancer is detected in its early stages so that they can receive timely therapeutic measures, such as surgical resection. Overall survival statistics may rise significantly as a result of this.

Therefore, the most pressing need is the development of accurate and reliable diagnostic tools that can effectively differentiate between stages and focus on both early and multi-stage identification of gastric cancer.

The revolutionary potential of artificial intelligence (AI) in obtaining more precise and prompt cancer diagnoses has been demonstrated by recent developments in deep learning. AI systems can learn to identify complex patterns and subtle abnormalities that even seasoned clinicians might overlook by utilizing large and varied medical datasets. Particularly in the field of gastric cancer, artificial intelligence (AI) has shown significant promise in the analysis of histopathological slides and endoscopic images, supporting the early identification and diagnosis of cancers.

By employing neural networks, such as Convolutional Neural Networks (CNNs), to automatically extract significant features without requiring human input, deep learning has significantly improved image-based diagnostics. These systems have demonstrated remarkable outcomes, particularly when it comes to identifying and categorizing stomach lesions from medical images. In certain instances, they have even surpassed human experts in their field. There are numerous significant advantages to diagnosing gastric cancer using artificial intelligence (AI) and deep learning. These include the capacity to assist in medical settings with limited resources, quicker analysis, more accurate results, and a lighter workload for medical professionals. In addition to being simpler to automate, AI and deep learning technologies can accurately and promptly identify lesions, enabling patients to receive timely curative treatments and enjoy improved health outcomes.

Although recent studies on classifying gastric cancer using histopathological images have shown promising accuracy, two main challenges still limit their use in clinical settings: poor generalization across different datasets and limited understanding of how models make decisions. Our study focuses on addressing the first challenge. We use pretrained CNN models, specifically DenseNet121, and carefully fine-tune the deeper layers using a thorough hyperparameter tuning process. This transfer learning approach makes use of already learned visual features while adjusting the model to recognize the specific patterns found in early gastric cancer.

By applying careful data preparation, augmentation, and k-fold cross-validation, we achieved significantly higher and more stable validation accuracy compared to models trained from scratch. This demonstrates superior generalization across datasets. Even though it's still hard to understand how models work, our results show that fine-tuned pretrained CNNs are a good and quick way to find early-stage gastric cancer. This work also sets an environment for subsequent study that aims at rendering AI systems easier to utilize in clinical settings. One big problem with organizing gastric cancer patients in groups is that the datasets have distinct characteristics because hospitals use different staining methods, scanners, and patient backgrounds. Because of these differences, standard CNN models don't work well with all types of data. To tackle this problem, our research utilizes transfer learning by refining pretrained CNN models, particularly DenseNet121, on two

significant and diverse histopathology datasets: GasHisSDB and SEED. These datasets include different types of tissues and stages of cancer, which helps the model learn more general and flexible features. By using good preprocessing, data augmentation, and cross-validation, our method makes the model stronger and more adaptable to different clinical settings. This method solves two big problems that came up in earlier research: making the model more reliable and easier to understand for finding gastric cancer.

2. PROPOSED METHOD

To tackle the challenges of early detection gastric cancer, this research proposes using Deep Learning techniques, specifically Convolutional Neural Networks (CNNs), for the automated analysis of histopathological images. The process begins by collecting a diverse and high-quality dataset of annotated gastric cancer tissue samples. This dataset is pre-processed to standardize image resolution, remove noise, and normalize pixel values to ensure consistency and clarity. Data augmentation methods like rotation, flipping, and scaling are applied to expand the dataset and improve the model's ability to generalize to new, unseen data. In this study, we implement a CNN-based framework using transfer learning, where pretrained DenseNet121 model is fine-tuned to detect the specific histopathological features of gastric cancer. This model is trained on carefully curated datasets to recognize both subtle and clear cancer patterns within tissue samples. This approach enhances accuracy and efficiency over traditional diagnostic methods and improves generalization across varied clinical data. The training phase focuses on optimizing the model using binary or multi-class cross-entropy loss functions, with class weighting used to manage class imbalance, ensuring the model can accurately classify cancerous and non-cancerous tissues. Once trained, the model is evaluated using performance metrics such as accuracy, precision, recall, and AUC-ROC to confirm its reliability. Overall, this deep learning approach offers a valuable advancement in the early and accurate detection of gastric cancer, with the potential to improve diagnostic outcomes and patient survival rates.

3. LITERATURE REVIEW

Tang et al. [1] A real-time AI-assisted system for detecting early gastric cancer was developed and validated using multicentre retrospective data. The system integrates deep learning to analyze endoscopic images and demonstrated superior diagnostic accuracy over conventional methods. It also reduced interobserver variability. The study's multicentre design strengthens its clinical generalizability. The findings suggest practical utility for early screening and diagnosis. This work highlights AI's transformative potential in real-time clinical workflows.

Sharma et al. [2] This study utilizes deep convolutional neural networks (DCNNs) to classify gastric carcinoma from whole-slide histopathology images. The method achieved high accuracy in distinguishing cancer subtypes. Feature extraction and spatial resolution played key roles in performance. The paper demonstrates how digital pathology combined with AI can automate cancer diagnostics. It contributes to reducing interobserver variability. The approach sets a precedent for large-scale image-based screening.

Ai et al. [3] This state-of-the-art review details current approaches in gastric histopathology image analysis using deep learning. It covers segmentation, classification, and detection techniques. The authors analyse trends in dataset use, model architecture, and clinical translation. Limitations such as lack of large public datasets and standard benchmarks are discussed. The review also explores explainability and generalizability challenges. It is a key reference for researchers building AI tools in gastric cancer diagnostics.

Jang et al. [4] A deep learning framework is introduced for the subclassification of gastric carcinoma using whole-slide images. The model differentiates histological subtypes with high precision. Class imbalance was addressed using advanced augmentation techniques. Subtyping is crucial for prognosis and treatment planning. The system provides consistent, reproducible classifications, reducing manual workload. The study paves the way for AI-assisted histological typing in clinical settings.

Hinata et al. [5] This study focuses on detecting immunotherapy-sensitive subtypes of gastric cancer using histologic image-based deep learning. The model identifies microsatellite instability (MSI) status from H&E-stained slides. It demonstrates that visual patterns associated with MSI can be learned by CNNs. This approach supports non-invasive patient stratification. The study is relevant for personalized treatment planning. It integrates histopathology with predictive oncology.

Wang et al. [6] The researchers developed a deep learning model to predict gastric cancer outcomes using histopathology images of resected lymph nodes. The model successfully forecasted recurrence risk and survival. It captured spatial and morphological patterns that correlate with prognosis. The tool offers support for post-operative treatment planning. It demonstrates the utility of AI in prognostication. The work bridges image analysis with survival prediction.

Kuntz et al. [7] This systematic review examines deep learning in gastrointestinal cancer histology, including gastric cancer. It categorizes studies by task type (classification, segmentation, prediction) and summarizes model performance. The review identifies gaps in dataset diversity and validation protocols. It emphasizes the need for transparent and reproducible research. The paper serves as a valuable synthesis of current knowledge. It informs best practices in model development and evaluation.

Ba et al. [8] This study assesses the effect of deep learning assistance on the diagnostic accuracy of pathologists. Using AI support, diagnostic accuracy and interobserver agreement improved significantly. The model acted as a second reader, especially helpful in ambiguous cases. The research validates AI's role in augmenting human performance. It also underscores the importance of clinician–AI collaboration. The study supports AI adoption in routine pathology.

Tung et al. [9] A deep learning method was developed to identify pathological slices in gastric cancer histopathology. The model filters out non-informative slices, streamlining diagnosis. It reduces the pathologist's workload while maintaining high accuracy. The system integrates into digital pathology workflows. It highlights pre-processing as a key step in AI pipelines. The study contributes to efficient clinical deployment of AI tools.

Wang et al. [10] This paper presents DEMoS, an ensemble deep learning framework for predicting molecular subtypes of gastric adenocarcinoma. It uses histopathological images to infer subtypes such as MSI, EBV-positive, and genomically stable. The system enhances patient stratification for targeted therapies. High performance across subtypes was demonstrated. This research links morphology with molecular phenotypes. It's a step toward AI-driven precision oncology. Zheng et al. [11] A deep learning model is proposed for predicting EBV-associated gastric cancer, combining human

expertise and AI. The system uses whole-slide images to infer EBV status, validated against clinical labels. Human–AI collaboration improved diagnostic performance. This approach reduces reliance on molecular assays. The study supports scalable, cost-effective EBV screening. It highlights synergy between computational and clinical expertise.

Yong et al. [12] This study uses deep ensemble learning on the GasHisSDB dataset to detect gastric cancer from histopathological images. The ensemble model outperformed individual CNNs. The approach enhances generalization and robustness. Evaluation metrics show high diagnostic accuracy. The paper validates the usefulness of GasHisSDB for model training. It exemplifies ensemble methods in digital pathology. .Several studies emphasized the importance of large, diverse datasets (e.g., GasHisSDB, SEED) and fine-tuning pretrained models to improve generalization across clinical environments. Reviews further highlighted challenges such as dataset heterogeneity, lack of interpretability, and the need for standardized validation protocols. Innovations like ensemble learning, visual transformers, and AI-human collaboration have shown promise in improving diagnostic accuracy and clinical utility.

Yoon et al. [13] developed a lesion-based CNN model to improve endoscopic detection and depth prediction of early gastric cancer (EGC). Trained on annotated images, the model outperformed expert endoscopists in sensitivity and specificity, demonstrating its potential to aid non-experts and support therapeutic decision-making. The study highlights the clinical value of AI in enhancing gastric cancer diagnostics. A comprehensive review of AI applications in pathological diagnosis of gastric cancer is presented. It discusses image preprocessing, model training, and validation approaches across studies. Various architectures, including CNNs and hybrid models, are evaluated. The paper emphasizes the role of AI in early diagnosis and risk stratification. It also explores challenges in clinical implementation such as data heterogeneity. The review serves as a roadmap for future AI developments in gastric pathology.

Deng et al. [14] A comprehensive review of AI applications in pathological diagnosis of gastric cancer is presented. It discusses image preprocessing, model training, and validation approaches across studies. Various architectures, including CNNs and hybrid models, are evaluated. The paper emphasizes the role of AI in early diagnosis and risk stratification. It also explores challenges in clinical implementation such as data heterogeneity. The review serves as a roadmap for future AI developments in gastric pathology.

Raghav Sundar et al. [15] emphasize that early detection of gastric carcinoma remains the most decisive factor in improving patient survival. Their work draws attention to serum-based biomarkers, molecular signatures, and genomic alterations as potential diagnostic tools for identifying early-stage disease. They also stress the value of integrating proteomic and computational approaches to refine risk stratification and improve diagnostic precision. Collectively, their studies suggest that advances in biomarker-driven strategies hold strong promise for transforming early gastric carcinoma detection.

4. METHODOLOGY

The objective of this methodology is to develop a high-performing deep learning system for the early detection of gastric cancer through histopathological image analysis. The gastric cancer dataset, which has Normal, Stage 1, and Stage 2 images, underwent a thorough data cleaning process to get rid of samples that were broken or had the incorrect labels. After cleaning, data augmentation methods like rotation, flipping, zooming, and contrast adjustment were used to make the data more diverse and fix the class imbalance, especially for the Stage 1 (early-stage carcinoma) images that weren't represented enough. To make sure the evaluation was fair, the dataset was split into 80% for training and 20% for validation. We used transfer learning to fine-tune the CNN model DenseNet121. Hyperparameter tuning was used to identify the best number of dense layers, learning rate, dropout rate, and optimizer for each model in order to improve performance. The main goal was to improve Stage 1 detection. The models were evaluated using standard metrics including accuracy, precision, recall, F1-score, and validationloss to ensure reliable and sensitive classification. The best-performing models showed high sensitivity and good generalization for detecting early-stage gastric cancer. The final predictive model was saved in HDF5 (.h5) format using Keras. During prediction, the input images are pre-processed, features are extracted, and the system generates classification results to indicate both the presence and stage of cancer

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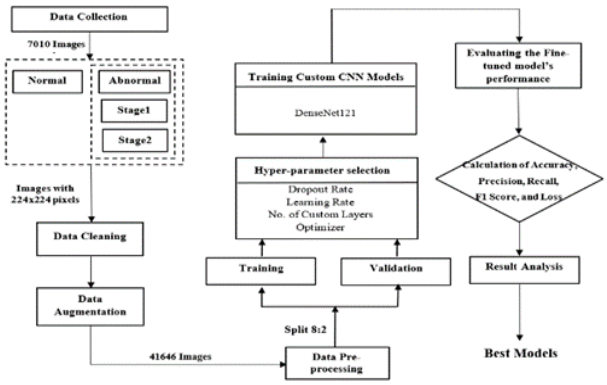


Figure 1: Workflow illustrating the process of gastric cancer classification.

4.1 Data Collection

In this study, we utilized three complementary sources of histopathological images. Two of these were public datasets: the GasHisSDB (Gastric Histopathology Subtype Database) and the SEED Gastric Carcinoma Dataset. GasHisSDB contains H&E-stained gastric tissue image patches labeled as either normal or adenocarcinoma (well-, moderately-, or poorly differentiated), available in three resolutions: 80×80, 120×120, and 160×160 pixels, and curated by medical professionals. The SEED dataset includes high-resolution images that cover various histological grades and tissue types, offering detailed visual data for both normal and cancerous gastric tissues. Additionally, we included a set of expert-annotated images obtained in collaboration with a gastroenterologist, helping to expand the representation of early-stage cancer lesions. In total, we curated 7,010 images, which were categorized into Normal, Stage 1 (early-stage carcinoma), and Stage 2 (progressive-stage carcinoma) classes. All images were resized to a uniform resolution of 224 × 224 pixels to standardize inputs for CNN-based analysis.

4.2 Pre-processing

The pre-processing stage prepares the images for model training by performing steps such as resizing, reshaping, and

converting them into numerical arrays. Since raw image files cannot be directly understood by computers, we use NumPy to convert each image into an array of pixel values, usually ranging from 0 to 255. These steps ensure that all input images have consistent dimensions and formats, which is essential before feeding them into the neural network for training.

4.3 Data augmentation

The heading for subsubsections should be in Times New Roman 11-point italic with initial letters capitalized. To prepare the images, their pixel values were normalized to a range between 0 and 1 by scaling them with a factor of 1/255. To enhance the model's ability to generalize and perform well in complex environments, eight different data augmentation techniques were used. This included rotation, width shift, height shift, and fill operations as illustrated in Figure 1.1. In addition, random horizontal flipping, shearing (up to 0.2 radians), and zooming (up to 20%) were applied to further diversify the training data. As a result, the augmentation process generated an additional 41,646 images, ensuring better and more uniform representation across all stages of gastric cancer[31].

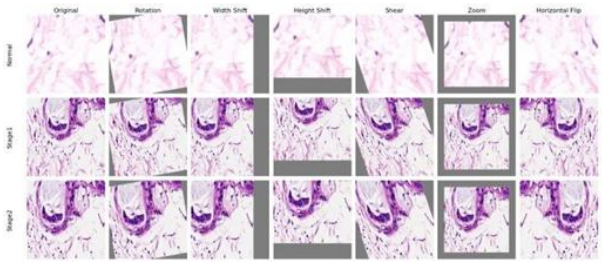


Figure 2: Data augmentation techniques applied to gastric cancer images

4.4 Hyperparameter Tuning

To improve model performance, Bayesian optimization was used through Keras Tuner to smartly search for the best set of hyperparameters [32]. A custom HyperModel class was created to include both the model architecture and the tunable parameters, enabling more flexible design and easier experimentation.The tuning process involved 25 trials per model across eight different CNN variants, providing a good balance between computational cost and thorough parameter exploration. Key hyperparameters included the number of convolutional layers, which ranged from 3 to 7, each with 512 filters, allowing the model to capture features at different levels of detail. The learning rate was adjusted between 0.0001 and 0.01, while dropout rates varied from 0.1 to 0.6, based on previous research and initial test results. These hyperparameters played a crucial role in determining the model's capacity, training speed, and its ability to generalize without overfitting. This careful tuning process led to better validation accuracy and overall model performance, all while keeping computational use efficient.

4.5 Experimental setup

The experiments were carried out on a system equipped with 16GB of RAM and an NVIDIA RTX 2050 GPU with 11GB of memory. This computational setup supported advanced methods such as Bayesian optimization for hyperparameter tuning and K-fold cross-validation, all without sacrificing speed or performance. The trained models were saved in .h5 format (Hierarchical Data Format)—one version captured after the initial training phase and another after fine-tuning. This was done to ensure reproducibility and allow for further experimentation in future studies

5. RESULTS AND DISCUSSIONS

A balanced dataset of gastric histopathology images—including normal, Stage 1, and Stage 2 samples—was used to evaluate DenseNet121 (CNN model), with the aim of identifying the best model for early-stage gastric cancer detection. Early results show that CNN-based methods not only improve diagnostic accuracy but also significantly reduce the time and manual effort involved in traditional pathology workflows. This study highlights the powerful role of deep learning in advancing early cancer diagnosis, especially in environments with limited access to medical experts.

Hyperparameter tuning played a key role in boosting the classification accuracy of CNN model for early-stage gastric cancer detection. A Bayesian optimization strategy was used to explore the best hyperparameter ranges, followed by K-fold cross-validation to ensure strong generalization and accurate performance evaluation [32]. DenseNet121 model was tuned to improve its ability to differentiate between normal, early-stage, and advanced-stage gastric cancer tissues.

The DenseNet121 model, using three custom dense layers, a learning rate of 0.0002, and a dropout rate of 0.4, showed strong performance. The RMSprop optimizer helped achieve stable training by adaptively adjusting learning rates based on gradient values which is providing a good balance between training speed and accuracy on the limited histopathology dataset.

Table 1:Optimal hyperparameter setting for Densenet121 model

Model/Best Parameter	No of Custom layers	Learning rate	Dropout Rate	Optimizer
DenseNet121	**5**	**0.0002**	**0.4**	**RMSprop**

As summarized in Table 1,it demonstrates that Densenet121 model can achieve competitive accuracy when appropriately tuned. This highlights the critical importance of customized hyperparameter optimization in developing high-performing deep learning systems for early-stage gastric cancer detection.

Table 2 summarizes the validation performance of DenseNet121 CNN model tested for early-stage gastric cancer detection, highlighting notable differences in classification accuracy and overall model effectiveness. DenseNet121 achieved an accuracy of 85.8%, demonstrating its strength in extracting intricate histopathological patterns through its densely connected layers. DenseNet121 strike a good balance between accuracy and efficiency for early gastric cancer detection, it may require further tuning or architectural adaptation to improve generalization in small- scale medical imaging tasks.

Table 2: Validation metrics for CNN models I tumor Classification

Model\ Metrics	Accuracy	Precision	Recall	F1-Score	Log loss
DenseNet121	0.8582	0.8649	0.8582	0.8562	0.3193

Figure 3 illustrate training accuracy and loss trends across 25 hyperparameter tuning trials (50 epochs each). Most models exhibited early convergence, with minimal signs of overfitting or underfitting. Among all architectures, DenseNet121 and MobileNetV2 achieved the most consistent and high-performing results, demonstrating strong generalization and low validation loss across folds.The best-performing DenseNet121 model were selected based on lowest validation loss and highest accuracy. The superior performance of DenseNet121 is attributed to its densely connected architecture, which promotes effective gradient flow.

The left panel illustrates the accuracy vs. epoch curves, depicting the progression of classification accuracy across 50 epochs for each model, thereby reflecting their learning behavior and convergence trends. The right panel presents the loss vs. epoch curves, highlighting the corresponding reduction in loss values, which serves as an indicator of each model's training effectiveness and optimization stability.

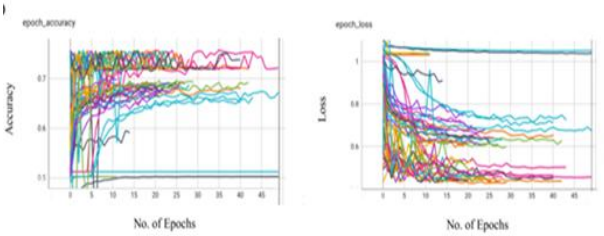


Figure 3: Performance metrics of CNN models for Gastric cancer classification through transfer learning.

The confusion matrix provides a comprehensive evaluation of classification performance by detailing the distribution of true positives, false positives, true negatives, and false negatives. This analysis is essential for understanding the accuracy, sensitivity, and specificity of each model. Figure 4 presents the confusion matrices obtained during the testing phase for all eight fine-tuned models. Darker cell intensities correspond to a higher proportion of correct predictions, offering visual insight into each model's effectiveness in distinguishing between histopathological grades.

DenseNet121 demonstrated the best overall performance, achieving 98.5% recall for Stage I with 100% Normal class accuracy after fine-tuning. However, it showed a sharp drop in Stage II accuracy due to misclassification into Stage I.

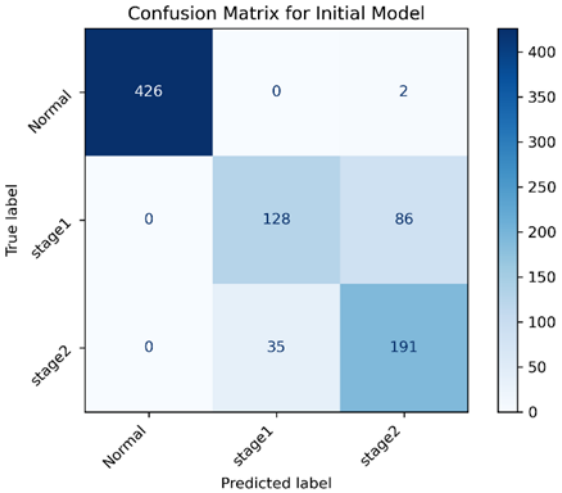


Figure 4:Confusion matrix for testing of fine-tuned model

Figure 5 presents the Receiver Operating Characteristic (ROC) curves for the fine-tuned CNN models, illustrating the trade-off between the true positive rate (TPR) and false positive rate (FPR) across varying classification thresholds. The Area Under the Curve (AUC) serves as a key performance metric, with values approaching 1.0 indicating stronger discriminative capability. For clarity and comparative analysis, ROC curves are displayed for the top eight models, enabling a comprehensive assessment of their effectiveness in early gastric cancer detection.

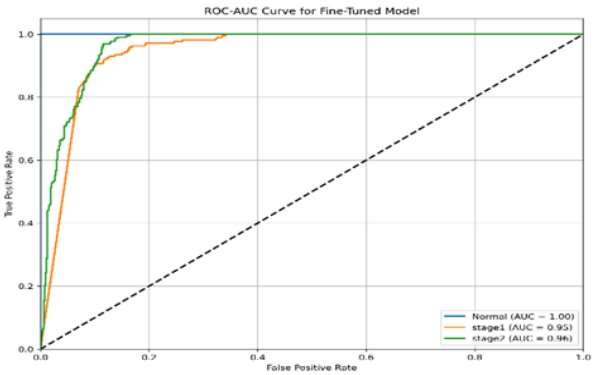


Figure 5: ROC-AUC curves depicting the classification performance for gastric cancer classification

Model shows AUC > 0.98 for Stage I, confirming high sensitivity in detecting early-stage cancer also demonstrates near-perfect classification, making them strong candidates for diagnostic support

6. CONCLUSION

This study demonstrates the feasibility and effectiveness of deep learning—particularly Convolutional Neural Networks (CNNs)—for automating the early detection and classification of gastric carncer. Traditional diagnostic methods are often labor-intensive, require specialized expertise, and may lack consistency, particularly in identifying early-stage cancer. By evaluating DenseNet121 achieving a high classification accuracy of 85.8%.

DenseNet121, with its densely connected layers, optimized feature reuse and minimized redundancy, enabling precise extraction of complex histopathological features, especially in subtle early-stage lesions. Analysis of confusion matrices and AUC-ROC curves confirmed both models' ability to accurately classify normal, pre-cancerous, and early cancerous tissues, with minimal misclassification across all stages. These results highlight the potential of CNN-based deep learning systems as support tools in clinical pathology, improving the early detection of gastric cancer and enabling timely treatment planning. However, certain limitations—such as a modest dataset size and limited diversity in histological imaging—restrict the full generalizability of the models.

Future research should aim to expand and diversify the dataset by including samples from multiple institutions, staining protocols, and imaging conditions. Incorporating semi-supervised learning techniques, advanced augmentation, and Explainable AI tools (e.g., Grad-CAM, saliency maps) will enhance model transparency and clinical trust. Further work should also explore lightweight and mobile-compatible architectures for integration into real-time diagnostic workflows, especially in low-resource healthcare settings.

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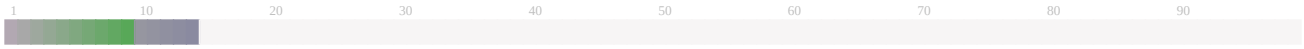
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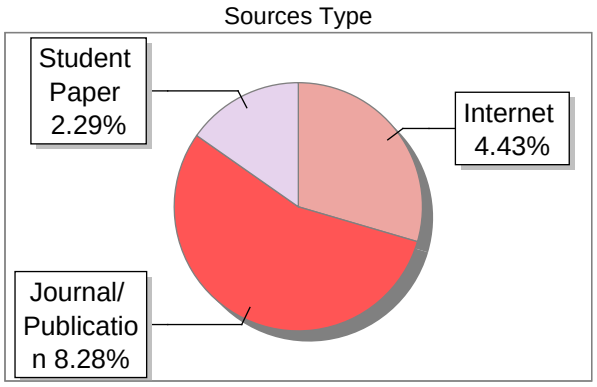
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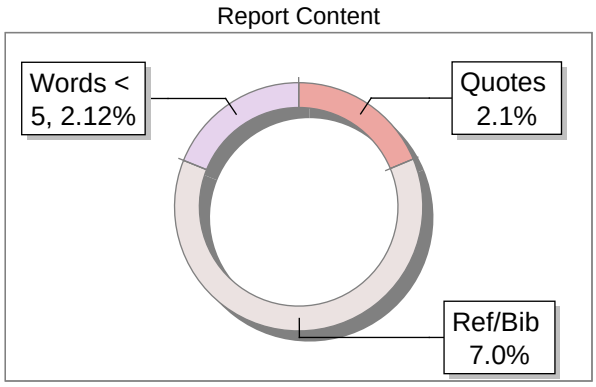
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